LGMM-Net: A Local–Global Encoder and Mask Mamba Decoder Network for Remote Sensing Change Detection

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***Abstract*—Remote sensing (RS) change detection (CD) is critical for applications such as urban planning and disaster assessment. It enables accurate identification of changes in ground objects across bi-temporal images. With the rapid development of deep learning, hybrid convolutional neural network (CNN)–transformer approaches have significantly advanced this field. However, existing fusion networks often lack robustness. They frequently misclassify pseudo-changes as true ones. Even with hard channel-splitting strategies that jointly model local and global features, this issue remains unresolved. To address these challenges, we propose a novel local–global encoder and mask Mamba decoder network (LGMM-Net). The network employs multikernel convolutions (multiconvs), window attention, and state-space sequence modeling to achieve robust fusion of local details and global semantics. First, we introduce a dynamic channel soft-splitting strategy within the dynamic local–global parallel encoder (DLGPE). This encoder integrates multiconvs, fixed-window attention, and shifted-window attention to capture both local details and global semantics. Second, we design the pyramid mask Mamba (PMM) module with a new scanning mechanism, termed mask-guided directional selective scan (MGD-SS2D). In this module, directional convolutions define scanning paths, while feature rotations enable four-directional modeling, thus enhancing boundary representation. Extensive experiments on four public datasets demonstrate that LGMM-Net consistently outperforms state-of-the-art (SOTA) methods in both accuracy and robustness. Our source code is available on https://github.com/Jhin1215/LGMM.**

***Index Terms*—Change detection (CD), Mamba, remote sensing (RS), state-space model (SSM), transformer.**

I. Introduction

REMOTE sensing (RS) change detection (CD) aims to identify dynamic surface changes by analyzing bi-temporal images of the same geographical region. This technology plays an indispensable role in urban management [1], [2], ecological resource monitoring [3], [4], and disaster emergency response [5], [6], providing critical data support for scientific decision-making.

Traditional CD methods are typically divided into three categories: arithmetic operation-based [1], [7], image transformation-based methods [8], [9], and classification-based methods [10], [11]. However, these approaches rely heavily on handcrafted features and are highly sensitive to illumination, atmospheric conditions, and seasonal variations, resulting in limited robustness. With the advancement of deep learning, convolutional neural network (CNN)-based models, including Siamese networks [12], U-Net [13] and their variants [14], [15], [16], [17], [18], [19] have significantly improved the accuracy of CD. Nonetheless, the inherent local receptive field of CNNs restricts their ability to capture global dependencies. To address this limitation, transformer-based models [20], [21], [22], [23], [24], [25], [26] have been introduced into CD, leveraging self-attention to reduce false alarms and missed-detection regions. Yet their quadratic computational complexity imposes substantial memory and runtime costs, limiting efficiency on large-scale RS images. Recently, Mamba [27], [28] has been adopted in vision tasks due to its linear complexity in state-space modeling and strong global sequence modeling capability. It has also been gradually applied to CD [29], [30], [31], [32]. However, existing Mamba-based approaches mainly emphasize unidirectional state transitions. They often neglect interchannel interactions and thus fail to capture fine-grained local details.

Despite remarkable progress in CD, existing methods still struggle to suppress pseudo-changes while retaining the ability to detect true changes. Bi-temporal images are often influenced by illumination, atmospheric conditions [33], and sensor parameters [34]. These factors introduce spectral or texture inconsistencies that appear as pseudo-changes. Also, seasonal factors such as vegetation growth, crop rotation, and snowmelt [35] can also introduce substantial background variations. For example, Fig. 1(a) illustrates the same land parcel (cyan box) transitioning from bare soil to vegetation. Although the land use is unchanged, the model incorrectly detects it as a true change. Pseudo-changes become even harder to handle when they interact with complex scene semantics. In Fig. 1(b), seasonal differences create drastically different backgrounds

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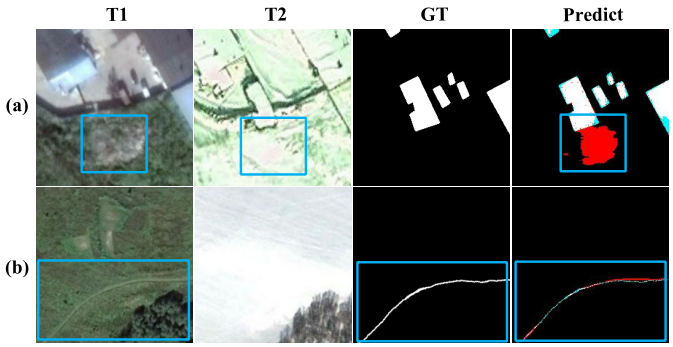


Fig. 1. Example images describe several unrelated changes. (a) Seasonal vegetation growth is misclassified as a true change. (b) Seasonal background shifts hinder the detection of subtle road changes. The first two columns show T1 and T2 images. The third column is the ground truth (GT) with changes in white and unchanged areas in black. The fourth column shows predictions, where red indicates false detections and cyan denotes missed-detection regions.

around roads, which severely obstructs the detection of subtle road changes. In this article, we regard these degradations, noise, and intertemporal variability as primary sources of pseudo-changes. Consequently, improving robustness to such variability is equivalent to reducing false alarms while preserving true changes. Several approaches have been proposed to address these challenges. ChangeFormer [22] mitigates pseudo-changes by aligning spatiotemporal background features through multilevel fusion and self-attention. BIT [21] encodes image features into semantic tokens and applies transformers to model global spatiotemporal dependencies. ELGC-Net [24] adopts channel hard-splitting to better fuse local and global semantics, thus enhancing discriminative ability. ConvFormer-CD [26] employs advanced spatiotemporal attention and feature fusion modules to align background features. However, these methods still face a fundamental tradeoff. They often suppress pseudo-changes at the cost of missing true ones, which limits precision–recall balance and constrains overall performance.

To systematically address the challenges of jointly modeling local details and global context, as well as accurately recovering change boundaries in CD, we propose a novel hybrid architecture named local–global encoder and mask Mamba decoder network (LGMM-Net). The model integrates parallel and collaborative designs in both the encoder and decoder. This enables it to efficiently capture local details, global semantics, and directional dependencies. First, we introduce the dynamic local–global parallel encoder (DLGPE) to overcome the limitations of conventional single-stream encoders in extracting both local and global features. DLGPE adopts a dynamic channel soft-splitting mechanism that adaptively assigns features to three parallel branches: multikernel convolution (multiconv), fine-grained fixed-window attention (fw_attn), and fine-grained shifted-window attention (sw_attn). This design allows simultaneous modeling of local and global information within the same layer. As a result, it significantly enhances the representation of complex geospatial scenes. Second, we propose the pyramid mask Mamba (PMM) module to reconstruct change boundaries with high precision from deep encoder features. Unlike conventional decoders that rely on simple upsampling and convolution, PMM treats decoding as a structure-aware reconstruction process. This approach effectively suppresses pseudo-changes and improves boundary completeness. PMM leverages the linear efficiency of the Mamba model for long-sequence processing. It also incorporates a novel mask-guided directional selective scan (MGD-SS2D) mechanism, where directional convolutions define scanning paths and feature rotations enable four-directional modeling. In addition, PMM aggregates pyramid features extracted by parallel dilated convolutions, explicitly modeling and restoring anisotropic objects. In summary, LGMM-Net introduces coordinated innovations in both the encoder and decoder. It combines the local perception of CNNs, the contextual modeling ability of transformers, and the long-sequence efficiency of Mamba. Together, these components provide a powerful and efficient solution for CD.

The main contributions of this work are summarized as follows.

1) We propose a novel hybrid CD framework, LGMM-Net, with a coordinated encoder–decoder design. It effectively integrates local details, global context, and directional dependencies, enabling efficient multiscale feature modeling.
2) We develop the DLGPE with an adaptive channel soft-splitting strategy. This design allows parallel modeling of local details and global semantics within a single layer. It also provides complementary enhancement, thereby improving the discriminative power of feature representations.
3) We design the PMM, a direction-aware multiscale component that incorporates a novel MGD-SS2D mechanism. It leverages directional masks to guide Mamba for efficient four-directional scanning. In addition, PMM aggregates multiscale dilated convolution features. Together, these designs enhance the representation of anisotropic objects and boundary regions while reducing computational cost.

The remainder of this article is organized as follows. Section II reviews related work in CD. Section III presents the details of LGMM-Net. Section IV describes the experimental settings and validates the effectiveness and efficiency of our model through quantitative metrics and qualitative analyses. Section V discusses the limitations and outlines future research directions. Section VI concludes the article.

II. Related Work

A. CNN-Based CD Methods

Owing to their strong capability in modeling local details, CNNs have been widely applied to CD. Daudt et al. [14] were the first to introduce CNN into CD tasks, proposing models such as FC-EF, FC-Siam-Conc, and FC-Siam-Diff. The FC-EF model adopts a single fully convolutional encoder–decoder architecture. Although simple, it struggles to capture spatiotemporal dependencies between bi-temporal images. In contrast, FC-Siam-Conc and FC-Siam-Diff, both based on the U-Net [13] architecture, employ Siamese

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encoders to extract bi-temporal features more effectively while reducing parameter complexity. These models have since become mainstream architectures in CD. Building upon UNet++ [36], Fang et al. [15] proposed SNUNet-CD, which leverages multilevel and compact skip connections to preserve fine spatial details and enhance contextual feature fusion. Similarly, Zhang et al. [16] introduced an attention mechanism to fuse multilevel original and difference features. They also applied deep supervision to optimize intermediate layers.

Nevertheless, CNN-based models are inherently limited by their local receptive fields. This limitation makes it difficult to capture long-range dependencies and global semantics. As a result, they often generate inaccurate boundary delineations and remain highly sensitive to pseudo-changes caused by illumination and seasonal variations. To mitigate these drawbacks, researchers have explored several optimization strategies. For example, Chen et al. [18] applied spatial and channel attention to capture long-range dependencies and enhance feature discrimination. This strategy reduces pseudo-changes caused by weakly discriminative features and class imbalance. Li et al. [17] proposed a lightweight CD network, A2Net, which mitigates the limited representational power of lightweight backbones by aggregating neighboring features and progressively integrating multilevel difference semantics. In addition, Ying et al. [37] provided the lightweight AEGL-Net, which combines adaptive multiscale enhancement with global–local fusion to improve robustness to degradation, noise, and temporal variability, while suppressing pseudo-changes caused by illumination and seasonal drift. Furthermore, Zhang et al. [19] developed an end-to-end superpixel-enhanced network to address the challenge of preserving object boundary integrity.

Recent works have incorporated attention mechanisms, deep supervision, boundary optimization, and lightweight designs. These strategies reduce false alarms and missed-detection regions to some extent. Nevertheless, CNNs remain fundamentally limited in their ability to capture global context. Consequently, when applied to complex RS scenarios with sparse changes or abundant pseudo-changes, CNN-based methods still leave substantial room for improvement.

B. Transformer-Based CD Methods

Benefiting from its inherent self-attention mechanism, the transformer architecture exhibits strong advantages in capturing long-range dependencies and modeling global semantics. Early research primarily focused on building pure transformer models. For example, Bandara and Patel [22] proposed a transformer-based Siamese network to efficiently extract and fuse multiscale change features. However, the quadratic complexity of standard global attention makes it difficult to apply to large-scale RS imagery. To address this issue, Liu et al. [38] introduced an efficient shifted-window self-attention mechanism, which maintains linear complexity while enabling cross-window information exchange. Inspired by this design, Zhang et al. [20] employed Swin transformer blocks to construct a full transformer-based Siamese U-Net, achieving end-to-end multiscale global feature extraction and change information recovery.

Although pure transformer architectures strengthen global context modeling, they often neglect the local details that are crucial for CD. To balance global dependencies and local details, hybrid CNN–transformer architectures have become the mainstream direction in CD. A representative paradigm uses CNNs as the backbone for feature extraction, followed by transformers for global relation modeling. For instance, Chen et al. [21] adopted CNN as the backbone to extract base features. They then condensed these features into tokens and applied transformer encoders and decoders for global context modeling. Liu et al. [23] proposed a CNN–transformer hybrid network to address long-range context loss and reduce the high computational costs of fine-grained farmland CD. Tang et al. [39] designed a dynamic local convolutional attention mechanism combined with a dual-decoder structure. This design effectively fuses multiscale information. Noman et al. [24] developed a dual-branch module, where one branch employed depthwise convolution and the other applied pooling-based self-attention. Together, they simultaneously capture local and global features. Wu et al. [40] leveraged extended long short-term memory (xLSTM) networks to model spatiotemporal relationships. This approach mitigates challenges in cross-temporal interaction and reduces feature redundancy. Similarly, Li et al. [41] integrated local feature extraction with global self-attention. They also incorporated multistage edge features to refine boundary precision. Ying et al. [37] further proposed a global–local parallel fusion framework that enhances global dependency modeling while reinforcing local feature representation with shared and context-aware weights.

Despite recent advances in hybrid CD models, most approaches still rely on predefined and fixed fusion strategies. This reliance limits their ability to adaptively balance local details and global semantics across diverse scenarios. To address this issue, we propose DLGPE, which integrates soft split with multiconv, fw_attn, and sw_attn. This design enables more comprehensive and flexible semantic modeling, thereby improving the model's generalization in complex and dynamic scenes.

C. Mamba-Based Methods

State-space models (SSMs) have attracted increasing attention due to their ability to efficiently model long sequences with linear complexity. Mamba [27] represents a significant advancement in this line of research by introducing a selective scanning mechanism that dynamically adapts model parameters to input content, thereby overcoming the content-insensitivity limitation of traditional SSMs. To extend Mamba to vision tasks, VMamba [28] generalizes 1-D sequence modeling to 2-D feature maps. It performs bidirectional scanning along horizontal and vertical axes, enabling effective global context aggregation and laying the foundation for Mamba-based vision models. Building on VMamba, Chen et al. [29] proposed ChangeMamba, which adapts selective scanning for CD by emphasizing relevant regions in multitemporal images. This design improves long-range dependency modeling across temporal phases and facilitates

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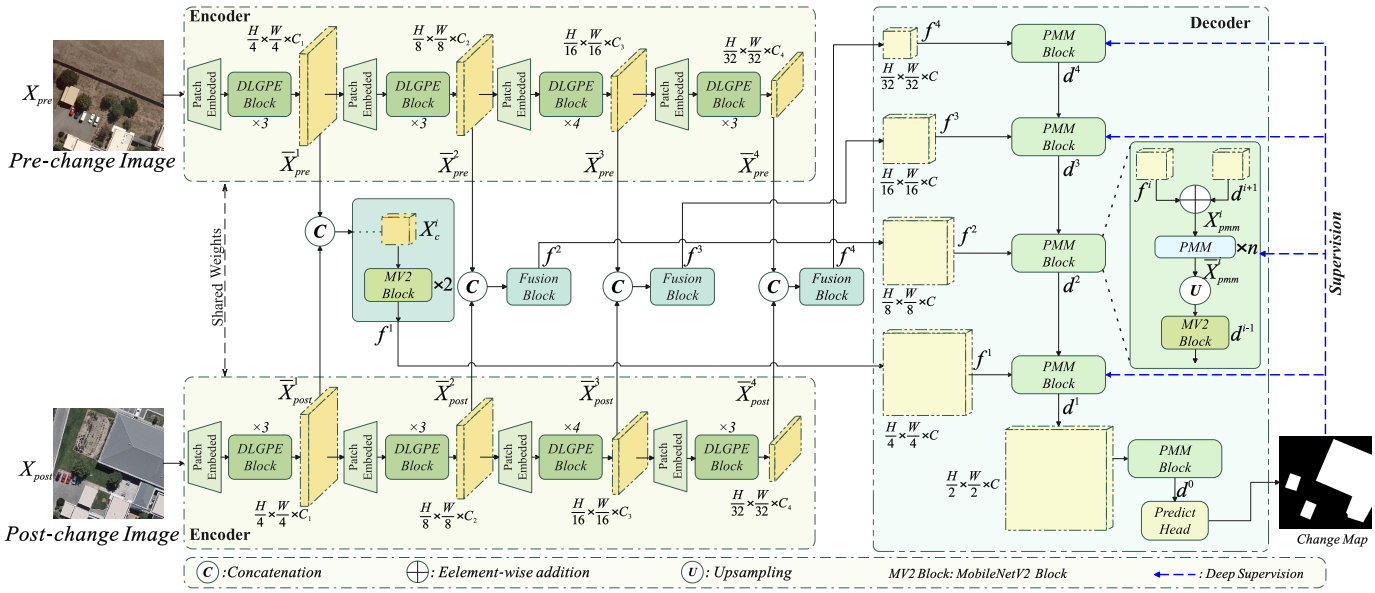


Fig. 2. Overall architecture of the proposed LGMM-Net. Given a bi-temporal image pair, a weight-sharing four-stage encoder built by stacking DLGPE blocks extracts multiscale representations at progressively reduced spatial resolutions. At each stage, the paired features are efficiently integrated by a lightweight fusion module based on MobileNetV2 inverted residual blocks (MV2 blocks), producing change-aware features for decoding. A five-layer PMM decoder then progressively upsamples and refines the features to predict the final change map. Intermediate prediction heads are used for deep supervision to stabilize multilevel learning.

more effective bi-temporal feature fusion. Zhao et al. [30] further extended the scanning strategy from four directions to eight, enhancing spatial representation capability. To compensate for insufficient local detail modeling, Zhang et al. [31] introduced a hybrid convolution–Mamba module that jointly captures local and global information. Zhao et al. [32] proposed ST-Mamba, which incorporates a spatiotemporal synergistic module (STSM) based on SSMs to explicitly capture spatiotemporal dependencies in multitemporal images. By unifying background representations across temporal phases, ST-Mamba suppresses pseudo-changes and improves feature fusion robustness.

Overall, existing Mamba-based CD methods mainly enhance spatiotemporal modeling by refining scanning strategies or combining Mamba with convolutional operations. While effective in capturing long-range dependencies, change inference is often performed implicitly, which may cause background variations such as illumination or seasonal changes to be misinterpreted as semantic changes. In contrast, PMM incorporates direction-aware long-range dependency modeling and multiscale local feature extraction within a unified decoding stage. Specifically, PMM employs a Mamba-based branch for directional long-range modeling and a parallel pyramid branch for multiscale feature extraction using dilated convolutions, whose outputs are linearly fused. This design offers a more explicit change modeling mechanism compared with existing Mamba-based approaches, without increasing scanning complexity.

III. Methods

In this section, we provide a detailed description of the proposed LGMM-Net. Section III-A outlines the overall network architecture. Sections III-B and III-C elaborate on the two core modules: the DLGPE and the PMM. Section III-D introduces the loss functions employed for model optimization.

A. Overall Architecture

The overall architecture of LGMM-Net is illustrated in Fig. 2. The network adopts a symmetric encoder–decoder design, consisting of a four-stage DLGPE, a fusion module, and a five-level multiscale decoder built with PMM blocks. The core concept lies in the collaborative synergy between the encoder and decoder, enabling the model to accurately capture genuine changes of ground objects. Moreover, it effectively suppresses pseudo-changes caused by illumination, seasonal, and sensor variability through three complementary designs: a novel adaptive feature routing strategy for robustness, cross-scale fusion with deep supervision for stable multilevel learning, and a novel direction-aware long-range modeling for boundary preservation.

In the encoding stage, bi-temporal images X_{pre} and X_{post} are fed into a weight-shared DLGPE encoder. Across four stages, multiple cascaded DLGPE blocks extract multiscale feature maps X^i_{pre} and X^i_{post} that are both robust to pseudo-changes and semantically rich, with spatial resolutions of $H/4 \times W/4$, $H/8 \times W/8$, $H/16 \times W/16$, and $H/32 \times W/32$. The detailed structure of DLGPE is described in Section III-B. At each scale, the feature maps from both bi-temporal inputs are passed into a fusion module composed of two stacked MobileNetV2 blocks, which perform spatiotemporal feature interaction to generate fused representations. This process can be formally expressed as

$$f^i = \text{Fusion}\left(X^i_{\text{pre}}\ X^i_{\text{post}}\right), \quad i \in \{1, 2, 3, 4\} \tag{1}$$

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where f^i denotes the fused features at the ith stage. In the decoding stage, the primary objective is to filter residual pseudo-changes while accurately reconstructing change boundaries. At the ith level, skip-connected encoder features are combined with the output of the previous decoder layer via element-wise addition, facilitating multilevel context aggregation. The fused features are then passed into a PMM block for deep decoding. To recover spatial resolution efficiently and avoid checkerboard artifacts, we employ PixelShuffle with an upscaling factor of two. Since this operation compresses the channel dimension, a pointwise (PW) convolution is applied to expand it fourfold, followed by a MobileNetV2 block for semantic enhancement. The final output of each level is the decoded feature representation. Notably, the deepest and shallowest layers lack corresponding fusion features and thus only accept single-stream inputs. The overall decoding process can be formulated as

$$d^i = \begin{cases} \text{PMM Block}\left(f^4\right), & i = 4 \\ \text{PMM Block}\left(f^i \oplus d^{i+1}\right), & i = 1, 2, 3 \\ \text{PMM Block}\left(d^1\right), & i = 0 \end{cases} \tag{2}$$

where $\oplus$ denotes the element-wise addition and d^i is the decoded feature at level i. The detailed design of the PMM block is presented in Section III-C. To ensure effective supervision across all levels, deep supervision is adopted. Specifically, the output of each decoder layer is fed into a lightweight prediction head (a PW convolution) to produce intermediate change maps

$$\hat{m}^i = \text{Head}\left(d^i\right), \quad i \in \{0, 1, 2, 3, 4\} \tag{3}$$

where $\hat{m}^0$ denotes the final prediction and the remaining $\hat{m}^i$ serve as auxiliary supervision signals during training.

B. Dynamic Local–Global Parallel Encoder

Traditional CD methods face an inherent tradeoff in feature encoding. On the one hand, CNN-based backbones are effective at capturing local textures and structural details of ground objects. However, their limited receptive fields restrict access to the global context. As a result, they struggle to distinguish true structural changes from pseudo-changes caused by illumination, seasonal variations, or atmospheric effects. On the other hand, transformer-based architectures are effective at modeling long-range dependencies. However, they often sacrifice the fidelity of local details. This limitation can lead to missed-detection regions of subtle or boundary changes.

To systematically address this local–global dilemma, we design the DLGPE. The key idea is to enable parallel processing with adaptive feature allocation within a single network stage, thereby jointly and efficiently modeling multidimensional features from bi-temporal images. Specifically, to handle the diversity of ground objects and the complexity of features in RS images, many CD methods adopt an equally divided channel strategy (hard split), such as ELGC-Net [24]. This strategy fixes a predefined ratio and partitions channels into local and global branches. It implicitly assumes a uniform balance between local textures and global semantics across layers and scenes. In practice, the optimal balance varies with object type, scale, illumination, and seasonal conditions. The resulting rigidity leads to suboptimal feature utilization and can amplify pseudo-changes. To address this issue, we propose a learnable dynamic channel soft-splitting mechanism (soft split). The mechanism predicts channel-wise weighting factors from input statistics and contextual cues, enabling scene-dependent feature routing across the three encoder branches. In this way, the encoder adaptively emphasizes local or global cues according to scene context and layer semantics, thereby improving feature discrimination while suppressing pseudo-changes.

The architecture of a DLGPE block is illustrated in Fig. 3(a). Within each block, the input multiscale features $X^i_{\text{pre/post}} \in \mathbb{R}^{H^i \times W^i \times C^i}$ are first normalized with LayerNorm making X^i_{in}, and then passed to the core module—the dynamic local–global parallel aggregation (DLGPA)—for parallel modeling of local and global representations, producing $X^i_{\text{out}} \in \mathbb{R}^{H^i \times W^i \times C^i}$. The output X^i_{out} is fused with the input through a residual connection, yielding $\overline{X}^i_{\text{out}}$. This result is further processed by another LayerNorm and a feed-forward network (FFN) for deep feature transformation, followed by a second residual connection with $\overline{X}^i_{\text{out}}$, which generates the final output $\overline{X}^i_{\text{pre/post}}$ of the block. DLGPA introduces several key innovations; the structural process is described as follows.

1) Dynamic Channel Soft-Splitting: We introduce a set of learnable parameters $\alpha^i \in \mathbb{R}^{H^i \times W^i \times 3}$ to guide the soft partitioning of channels. First, α^i is decomposed along the channel dimension into $\alpha_1 \in \mathbb{R}^{H^i \times W^i \times 1}$, $\alpha_2 \in \mathbb{R}^{H^i \times W^i \times 1}$, and $\alpha_3 \in \mathbb{R}^{H^i \times W^i \times 1}$. Each is then multiplied with the input features $X^i_{\text{in}} \in \mathbb{R}^{H^i \times W^i \times C^i}$ using the Hadamard product, producing weighted features for the three branches

$$\alpha_1, \alpha_2, \alpha_3 = \text{Split}\left(\alpha^i\right) \tag{4}$$

$$x^i_{\text{sw}}, x^i_l, x^i_{\text{fw}} = X^i_{\text{in}} \odot \alpha_1, X^i_{\text{in}} \odot \alpha_2, X^i_{\text{in}} \odot \alpha_3 \tag{5}$$

where $\odot$ denotes the Hadamard product. This design provides the model with high flexibility, enabling it to intelligently balance whether local textures or global semantics should be emphasized at each layer.

2) Fine-Grained Window Attention: To efficiently and differentially model global and semi-global context within DLGPA, we design two parallel and complementary branches: sw_attn and fw_attn.

Traditional window attention generates Q, K, and V through linear projections, which flatten the feature maps and break the inherent 2-D spatial priors of RS images. To address this issue, we employ two distinct convolutional operations to generate Q, K, and V in the two branches, ensuring that their feature extraction strategies align with their respective functional roles and yield more spatially meaningful attention maps.

In the sw_attn branch, we use 3×3 dilated convolutions with dilation $d = 2$ to generate $Q^i_{\text{sw}}, K^i_{\text{sw}}$, and V^i_{sw}

$$Q^i_{\text{sw}}, K^i_{\text{sw}}, V^i_{\text{sw}} = \text{Split}\left(\psi\left(X^i_{\text{sw}}\right)\right) \tag{6}$$

where ψ denotes a 3×3 dilated convolution ($d = 2$), and $X^i_{\text{sw}} \in \mathbb{R}^{H^i \times W^i \times C^i}$, $Q^i_{\text{sw}}, K^i_{\text{sw}}, V^i_{\text{sw}} \in \mathbb{R}^{H^i \times W^i \times \text{d}}$ with d representing the dimension of the attention subspace. This design preserves the 2-D structure while enlarging the receptive field to capture

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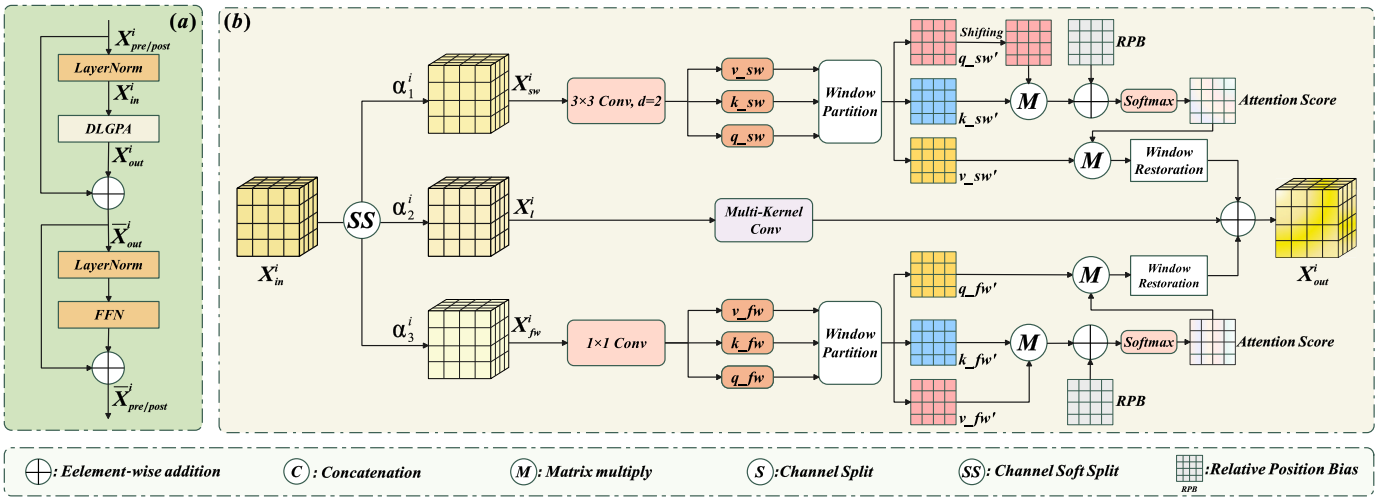


Fig. 3. (a) Illustration of the DLGPE. (b) Illustration of the DLGPA, which serves as the core component of DLGPE. DLGPA contains three parallel branches: sw_attn for global modeling, multiconv for local feature extraction, and fw_attn for semi-global modeling. The input features are first divided through channel-wise soft splitting, where three learnable parameters α_1^i, α_2^i, and α_3^i dynamically allocate channels to different branches. The sw_attn branch employs cyclic window shifting to capture long-range dependencies, the multiconv branch enhances detailed textures and boundaries, and the fw_attn branch strengthens mid-range contextual interactions. Finally, the outputs of the three branches are fused to form the aggregated representation.

long-range dependencies. To further suppress isolated pseudo-changes, we follow the Swin transformer idea but shift only Q cyclically (rather than both Q and K) to enable cross-window interactions. After window partitioning, cyclic shifting, and self-attention, we obtain

$$\mathrm{Attn}_{\mathrm{sw}}^i = \tilde{Q'}_{\mathrm{sw}}^i \times \sigma\left(K'_{\mathrm{sw}}^i \times V'_{\mathrm{sw}}^i\right) \tag{7}$$

$$\overline{\mathrm{Attn}}_{\mathrm{sw}}^i = \mathbf{R}\left(\mathrm{Attn}_{\mathrm{sw}}^i, H^i, W^i\right) \tag{8}$$

where σ denotes Softmax, $\times$ denotes the matrix multiplication, and $\mathbf{R}$ restores the original windowed structure.

In the fw_attn branch, we adopt PW (1×1) convolutions to generate $Q_{\mathrm{fw}}^i, K_{\mathrm{fw}}^i$, and V_{fw}^i. Unlike dilated convolutions, PW convolutions do not aggregate spatial neighborhoods, but instead focus on efficient channel interactions and feature transformation. This allows fw_attn to model abstract regional semantics within each window, complementing the global spatial sensitivity of sw_attn. The attention process is defined as

$$\mathrm{Attn}_{\mathrm{fw}}^i = Q'_{\mathrm{fw}}^i \times \sigma\left(K'_{\mathrm{fw}}^i \times V'_{\mathrm{fw}}^i\right) \tag{9}$$

$$\overline{\mathrm{Attn}}_{\mathrm{fw}}^i = \mathbf{R}\left(\mathrm{Attn}_{\mathrm{fw}}^i, H^i, W^i\right) \tag{10}$$

where σ denotes Softmax, $\times$ denotes the matrix multiplication, and $\mathbf{R}$ restores the original windowed structure. Together, the two branches provide complementary context modeling, enabling DLGPA to jointly capture long-range dependencies and fine-grained regional semantics.

3) Multikernel Convolution: To ensure that the model captures fine-grained textures and boundary information—crucial for recognizing real changes—while still attending to the global context, we design a multiconv branch as a strong complement to local feature extraction. As shown in Fig. 4, the input feature $X_l^i \in \mathbb{R}^{H^i \times W^i \times C^i}$ is split along the channel dimension into n groups, denoted as $X_{l1}^i, X_{l2}^i, \ldots, X_{\mathrm{ln}}^i$. Each

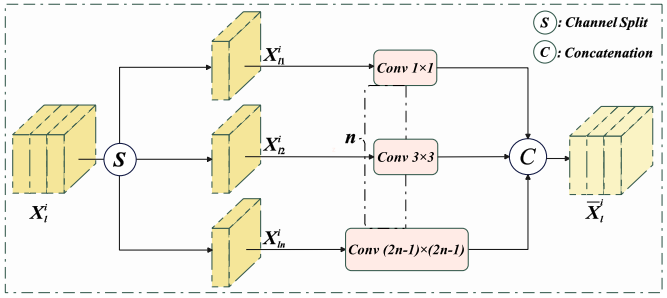


Fig. 4. Illustration of the multiconv. In the figure, n denotes the number of convolution kernels used in each encoder stage.

group is processed in parallel with a convolution kernel of different size, ranging from 1×1 to $(2n-1) \times (2n-1)$

$$\begin{cases} \tilde{X}_{l1}^i = \psi\left(X_{l1}^i\right) \\ \tilde{X}_{l2}^i = \psi\left(X_{l2}^i\right) \\ \cdots \\ \tilde{X}_{\mathrm{ln}}^i = \psi\left(X_{\mathrm{ln}}^i\right) \end{cases} \tag{11}$$

where ψ denotes a convolution operation with kernel size $2n-1$. The outputs of all branches are concatenated to form a comprehensive, multiscale local feature representation $\overline{X}_l^i$. This design enables the model to simultaneously capture high-frequency details (e.g., edges and textures) and mid-scale structures (e.g., object contours) within the same layer.

Finally, the outputs of the three parallel branches—sw_attn, multiconv, and fw_attn—are aggregated via element-wise addition. Since the dynamic channel partition preserves the complete channel dimension, this direct summation efficiently fuses complementary information, including global dependencies, multiscale local details, and semi-global context, yielding the final output feature X_{out}^i

$$X_{\mathrm{out}}^i = \mathcal{SW}\left(X_{\mathrm{sw}}^i\right) \oplus \mathcal{FW}\left(X_{\mathrm{fw}}^i\right) \oplus \mathcal{M}\left(X_l^i\right) \tag{12}$$

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where $\oplus$ denotes element-wise addition. $\mathcal{SW}$, $\mathcal{FW}$, and $\mathcal{M}$ represent the complete operations of the sw_attn, fw_attn, and multiconv, respectively.

In summary, the proposed DLGPE effectively suppresses pseudo-changes caused by illumination, seasonal, and sensor variability by reweighting channels in a data-dependent manner before branch aggregation. Moreover, it balances local detail preservation and global context modeling. By adaptively routing features across the sw_attn, fw_attn, and multiconv branches, it achieves a fine synergy between CNN and transformer representations, thereby enhancing change discrimination and reducing pseudo-change responses.

C. Pyramid Mask Mamba

Having extracted robust multiscale features via DLGPE, we now introduce the PMM as the decoder core to reconstruct accurate change maps. Mamba is effective in modeling the global context. However, existing Mamba-based CD methods still face several challenges in RS imagery. These methods typically rely on multiple or multidirectional scans to capture complex dependencies, which introduce heavy computational overhead and hardware bottlenecks. They also increase the risk of overfitting. More critically, the scanning mechanism often fails to restore the boundaries of directional man-made objects, e.g., buildings and roads, resulting in blurred contours, and false alarms.

To address the above-mentioned issues, we propose a novel MGD-SS2D mechanism that enables the model to efficiently capture anisotropic features during long-range spatial modeling. Convolution can be interpreted as an approximation of the iterative process in a SSM. A standard convolution with stride 1 can be expressed as

$$Y = X * K \tag{13}$$

where $K \in \mathbb{R}^{k\times k}$ is the kernel and $X \in \mathbb{R}^{H\times W}$ denotes the feature map. The output at location (i, j) is a linear combination of its neighborhood

$$y_{i,j} = \sum_{m=0}^{k}\sum_{n=0}^{k} x_{i+m,j+n}\cdot k_{m,n} \tag{14}$$

where $k_{m,n}$ is the kernel weight at position (m, n). This process is conceptually similar to the hidden-state update in an SSM

$$h_{i,j} = A \odot h_{\bar{i},\bar{j}} + B \odot x_{i,j} \tag{15}$$

$$y_{i,j} = C \odot h_{i,j} \tag{16}$$

where $\odot$ denotes the Hadamard product, h is the hidden state, $h_{\bar{i},\bar{j}}$ its predecessor, and A, B, and C are the transition, input, and output matrices, respectively. These matrices are effectively encoded within the convolution kernel K.

Although a standard kernel provides multiangle receptive fields, it remains isotropic within its $k \times k$ neighborhood. In other words, it assigns homogeneous weights to all pixels, ignoring directionality. This isotropy limits its ability to distinguish directional changes, e.g., building edges from nonstructured variations. Therefore, standard kernels are suboptimal for CD tasks requiring directional sensitivity.

To overcome this limitation, we introduce a kernel initialization strategy that embeds strong directional priors before scanning. As shown in Fig. 5(c), for a DWConv kernel $K \in \mathbb{R}^{C_1\times C_2\times k\times k}$, we first define a directional mask mask $\in \mathbb{R}^{k\times k}$ with values decaying along one axis. We then reshape it to mask $\in \mathbb{R}^{1\times 1\times k\times k}$ and apply the Hadamard product with K

$$\text{MaskedDWConv} = \text{mask} \odot K \tag{17}$$

where $\odot$ denotes the Hadamard product. This operation produces a kernel that responds more strongly to features along specific directions, thereby enhancing the discrimination between directional changes and nondirectional noise.

As illustrated in Fig. 5(b), we further design an efficient four-directional scanning mechanism consisting of forward and backward passes along both the horizontal and vertical axes. To avoid defining four independent kernels, we construct only two MaskedDWConvs, horizontal and vertical, and simulate the reverse scans through a simple flip–convolve–flip operation. Specifically, let $\mathcal{F}_h(\cdot)$ and $\mathcal{F}_v(\cdot)$ denote horizontal and vertical flip operations, and let $\text{DWConv}_h(\cdot)$ and $\text{DWConv}_v(\cdot)$ represent the corresponding directional convolutions. The forward scans are directly applied to the input feature X, whereas the reverse scans are obtained by flipping the input, performing the same convolution, and flipping the result back. For example, the horizontal reverse scan can be formulated as

$$X_{\text{out}}^{(3)} = \mathcal{F}_h^{-1}\left(\text{DWConv}_h\left(\mathcal{F}_h\left(X\right)\right)\right) \tag{18}$$

where $\mathcal{F}_h^{-1}$ is identical to $\mathcal{F}_h$ since the flip operation is self-inverse. The vertical reverse scan can be formulated in the same way using $\mathcal{F}_v$. This strategy efficiently generates four-directional responses, covering both forward and backward passes along horizontal and vertical directions. It introduces no additional parameters or computational cost and preserves a consistent receptive-field orientation. Finally, four learnable weights $\varepsilon_1, \ldots, \varepsilon_4$ are used to adaptively combine the directional outputs X_{out}^{j}

$$\overline{X}_{\text{mgd}}^{i} = \sum_{j=1}^{4} \varepsilon_j \odot X_{\text{out}}^{j} \tag{19}$$

where $\odot$ denotes the Hadamard product. This weighted aggregation enables the model to dynamically emphasize contextual cues from different directions.

Beyond directional modeling, computational efficiency is also considered. Traditional vision Mamba models (e.g., Vim) often require iterative or sequential unfolding when processing 2-D data, leading to high complexity and hardware bottlenecks. In contrast, our MGD-SS2D integrates SSM computations with convolution, enabling fully parallelized operations. This design dramatically reduces scanning complexity, allowing deeper decoders that recover boundaries more accurately and suppress false changes.

Based on the above analysis, we embed the proposed MGD-SS2D as the core of a U-Net-style multiscale decoder, termed PMM, to efficiently recover anisotropic object boundaries during decoding. As shown in Fig. 5(a), PMM adopts a dual-branch architecture that jointly decodes input features. The

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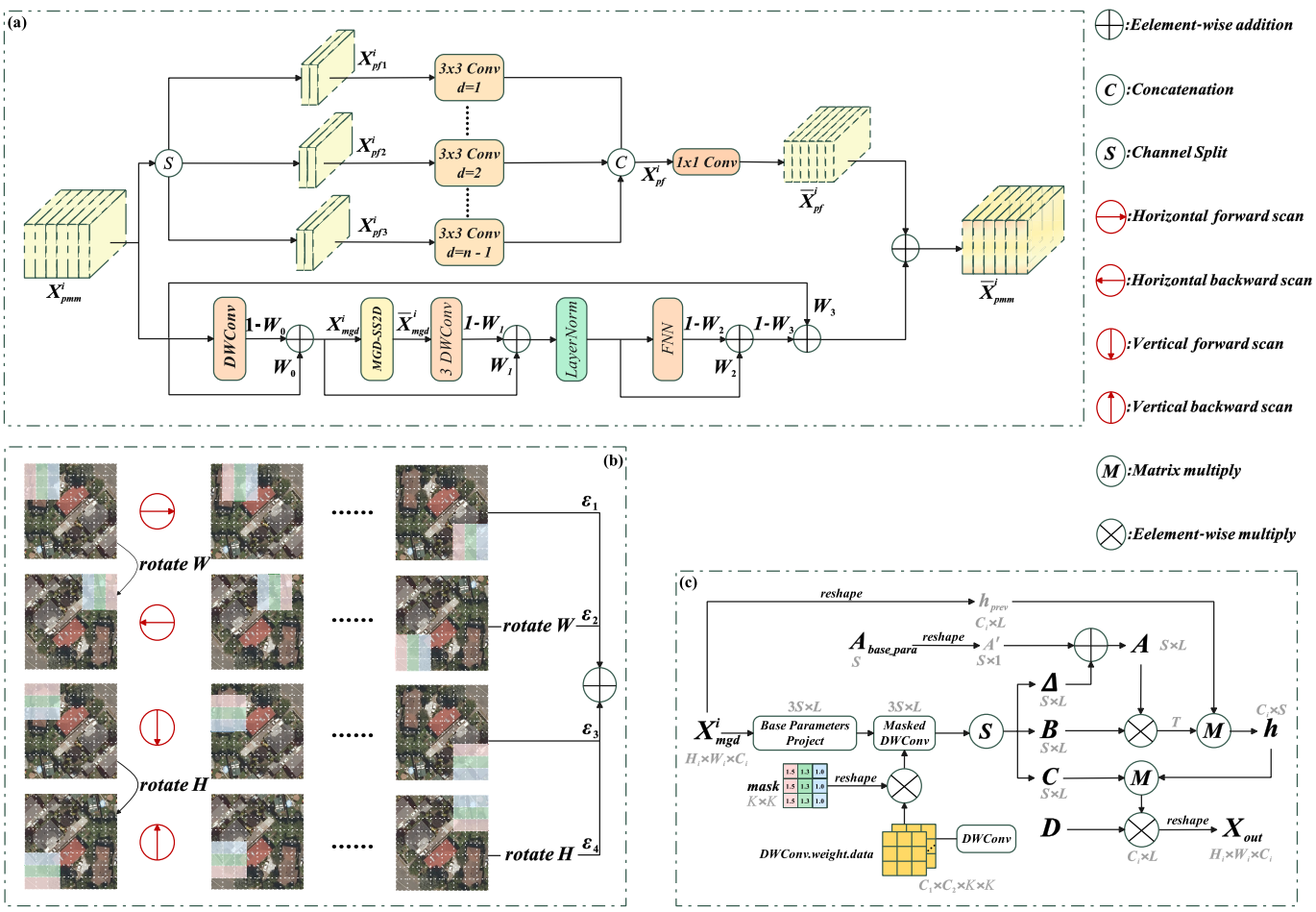


Fig. 5. Overview of the proposed PMM and the MGD-SS2D mechanism. (a) Structure of the PMM block. A dual-branch decoder integrates direction-aware long-range dependencies with multiscale local context. Learnable parameters W_0, W_1, W_2, and W_3 regulate dynamic residual connections. (b) Illustration of MGD-SS2D. It supports efficient four-directional scanning along the horizontal and vertical axes, in both forward and backward directions, using only two masked depthwise convolutions (MaskedDWConvs). Reverse scans are implemented via spatial flipping. The four-directional outputs are adaptively fused with learnable weights ε_1, ε_2, ε_3, and ε_4. Rotate W and Rotate H denote a 180° flip along the width and height dimensions of the feature map, respectively. (c) Kernel initialization and parallelized pipeline. A directional mask is applied to DWConv kernels to encode anisotropic priors. The masked kernels produce state-space parameters, enabling fully parallel computation to reduce complexity and improve boundary recovery.

long-range dependency and direction-aware branch builds on MGD-SS2D. Given input features $X^i_{\text{pmm}} \in \mathbb{R}^{H^i \times W^i \times C^i}$, we first apply a 3×3 DWConv to extract semantic features. A learnable weight W_0 fuses them with the raw input via an adaptive residual connection. The fused feature is then processed by MGD-SS2D. Its output passes through another 3×3 DWConv and a second residual fusion with W_1. We apply LayerNorm for stable training, and an FFN enhances nonlinearity, balanced by W_2. Finally, the branch output connects back to the original input X^i_{pmm} through W_3. This residual design captures long-range dependencies and explicitly models directional variation.

The multiscale local context branch provides complementary fine-grained details. It preserves small objects that may vanish during global modeling. The input X^i_{pmm} is split along the channel dimension to form a pyramid $X^i_{\text{pf1}}, X^i_{\text{pf2}}, \ldots, X^i_{\text{pfn}}$. Each pyramid level applies a 3×3 dilated convolution with dilation rates $d = 1, \ldots, n$. This design captures multiscale context without extra cost. The outputs are concatenated and linearly combined by a PW convolution, producing $\overline{X}^i_{\text{pf}}$.

Finally, we aggregate the two branches via element-wise addition. This fusion integrates direction-aware long-range dependencies with multiscale local details, yielding the decoded feature map $\overline{X}^i_{\text{pmm}} \in \mathbb{R}^{H^i \times W^i \times C^i}$.

The pipeline of the MGD-SS2D scanning mechanism is illustrated in Fig. 5(c). Given input $X^i_{\text{mgd}} \in \mathbb{R}^{H^i \times W^i \times C^i}$, we reshape it into a sequence $\mathbb{R}^{C^i \times L}$ with $L = H^i \times W^i$. A shared 3×3 DWConv extracts base parameters $p \in \mathbb{R}^{3S \times L}$. For each scan direction, the corresponding MaskedDWConv generates the SSM parameters

$$B, C, \Delta = \text{Split}\left(\text{MaskedDWConv}\left(p\right)\right). \tag{20}$$

We construct the transition matrix A by combining a static prior base_para $\in \mathbb{R}^S$ with a dynamic component Δ through element-wise addition. The resulting matrix A is then multiplied element-wise with B to produce a semantic selection matrix M, which controls the decay of past states and the weighting of new inputs. The hidden state is updated by multiplying M with the previous state h_{prev}. This updated hidden state is subsequently projected by C to generate the output

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sequence X_{seq}. Finally, a learnable scalar D modulates the directional contribution through element-wise multiplication with X_{seq}, followed by a reshaping operation. This design enables efficient, direction-aware scanning in a single parallel step.

In summary, the proposed MGD-SS2D effectively unifies directional sensitivity and parallel efficiency within a single formulation. By embedding it into the PMM decoder, our framework can recover fine-grained boundaries and suppress pseudo-changes with minimal computational cost. Consequently, PMM serves as a robust decoding component that bridges global context modeling and local detail preservation, leading to more reliable CD results across diverse RS scenarios. Together with the pyramid branch, the direction-aware MGD-SS2D delivers cross-scale, variability-robust decoding that maintains sharp boundaries while avoiding spurious responses from illumination and seasonal shifts.

D. Optimization Strategy

The optimization of LGMM-Net consists of a pixel-wise cross-entropy loss (CE Loss) and deep supervision. CD is essentially a binary classification task that determines whether each pixel belongs to the changed or unchanged class. The CE Loss is defined as

$$\mathcal{L}_{\text{CE}} = -\frac{1}{N}\sum_{i=1}^{N}\left[y_i \log(p_i) + (1-y_i)\log(1-p_i)\right] \tag{21}$$

where N is the total number of pixels, $y_i \in \{0,1\}$ denotes the GT label of pixel i (0 for unchanged and 1 for changed), and p_i is the predicted probability that pixel i belongs to the changed class.

To mitigate gradient vanishing in deep networks and ensure effective training of features from deep semantic layers to shallow details, we adopt a deep supervision strategy. As illustrated in Fig. 2, the output of each PMM Block in the decoder, $d^i (i = 0, 1, 2, 3, 4)$, is passed through a prediction head to generate a corresponding change map m^i. During training, each m^i is supervised against the GT. The final loss for backpropagation is the sum of these multiscale losses

$$\mathcal{L}_{\text{total}} = \sum_{i=0}^{4} \mathcal{L}\left(m^i, GT\right). \tag{22}$$

This design applies supervision signals directly to intermediate layers, encouraging hierarchical feature learning and enabling the model to produce change maps with sharper boundaries and richer details.

IV. Experiments and Results Analysis

In this section, we conduct a comprehensive evaluation of the proposed method on four widely used public CD datasets. Through extensive quantitative and qualitative analyses, we systematically demonstrate the effectiveness and superiority of LGMM-Net compared with SOTA methods.

A. Datasets

To comprehensively evaluate the performance of the proposed method across diverse scenarios, we conducted experiments on four widely used public CD datasets.

1) LEVIR-CD [42]: This is a large-scale building-CD dataset. It contains 637 pairs of high-resolution images (0.5 m/pixel) collected from Google Earth, with a temporal span ranging from 5 to 14 years. The dataset focuses on urban building changes. We cropped the images into nonoverlapping 256×256 patches and split them into 7120, 2048, and 1024 pairs for training, validation, and testing, respectively.

2) CDD [35]: This large-scale dataset contains 11 pairs of high-resolution multispectral images from Google Earth, with spatial resolutions ranging from 0.03 to 1 m. It not only captures changes in man-made objects (e.g., buildings, roads, and vehicles) but also introduces complex background variations caused by seasonal vegetation phenology. We cropped the images into nonoverlapping 256×256 patches and split them into 10 000, 3000, and 3000 pairs for training, validation, and testing, respectively.

3) GZ-CD [35]: This dataset consists of 19 pairs of high-resolution images (0.55 m/pixel) covering suburban Guangzhou, China, over a 13-year period (2006–2019). It records long-term land-cover and urban development changes. We cropped the images into nonoverlapping 256×256 patches and split them into 2834, 400, and 325 pairs for training, validation, and testing, respectively.

4) UAV-CD+ [43]: This dataset contains 2002 pairs of low-altitude UAV images (1024×1024, 0.06 m/pixel) captured in Guangdong Province, China, with a two-year interval between acquisitions. It covers diverse change types, including buildings, construction sites, land, and water areas. All images were cropped into nonoverlapping 256×256 patches and split into 25 626, 3203, and 3203 pairs for training, validation, and testing, respectively.

B. Assessment Metrics

To evaluate the performance of our model, we use six widely used metrics in CD: precision (Pre.), recall (Rec.), $F1$ score (F1), intersection over union (IoU), overall accuracy (OA), and Cohen's Kappa coefficient (κ). Among these, F1 and IoU are considered the core metrics, as they provide a more comprehensive assessment of model performance.

1) Precision (Pre.): It measures the proportion of true positives (TPs) among all samples predicted as changes

$$\text{Pre.} = \frac{\text{TP}}{\text{TP} + \text{FP}}. \tag{23}$$

2) Recall (Rec.): It measures the proportion of TPs among all actual changes

$$\text{Rec.} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{24}$$

3) F1 Score: The harmonic mean of precision and recall serves as the primary metric in CD tasks

$$F1 = \frac{2 \times \text{Pre.} \times \text{Rec.}}{\text{Pre.} + \text{Rec.}}. \tag{25}$$

4) Intersection Over Union: It quantifies the geometric overlap between predicted change regions and the GT changes

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}. \tag{26}$$

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TABLE I

COMPARISON WITH OTHER METHODS ON THREE PUBLIC DATASETS. THE BEST RESULTS ARE IN RED AND THE NEXT-BEST IN BLUE. ALL VALUES ARE IN %. MODELS ARE GROUPED ACCORDING TO THEIR PREDOMINANT ARCHITECTURAL CHARACTERISTICS. OUR METHOD IS CLASSIFIED AS MAMBA-BASED SINCE THE MAMBA DECODER CONSTITUTES THE CORE INNOVATION AND PLAYS A DECISIVE ROLE IN MODELING LONG-RANGE DEPENDENCIES. FOR FAIR COMPARISON, THE GROUPING FOLLOWS THE DOMINANT ARCHITECTURAL PARADIGM RATHER THAN THE INCLUSION OF AUXILIARY COMPONENTS. "/48" DENOTES THE CONVFORMER-CD VARIANT WITH A BASE CHANNEL WIDTH OF 48

Type	Method	LEVIR-CD						CDD						GZ-CD					
		F1	Rec.	Pre.	IoU	OA	Kappa	F1	Rec.	Pre.	IoU	OA	Kappa	F1	Rec.	Pre.	IoU	OA	Kappa
CNN-based	FC-EF [14]	86.31	84.81	87.85	75.91	98.63	85.58	73.10	60.65	91.97	57.60	94.73	70.31	69.89	57.59	88.86	53.71	95.55	67.60
	FC-Siam-Diff [14]	88.42	86.11	90.85	79.24	98.85	87.81	75.50	63.96	92.12	60.64	95.10	72.88	70.34	57.69	90.09	54.25	95.63	68.10
	FC-Siam-Conc [14]	89.29	88.15	90.45	80.64	98.92	88.72	74.63	64.35	88.81	59.52	94.84	71.84	70.45	58.13	89.40	54.38	95.62	68.20
Transformer-based	BIT [21]	90.25	88.35	92.23	82.23	99.03	89.74	94.46	93.78	95.15	89.51	98.70	93.73	81.73	77.08	86.98	69.11	96.91	80.05
	ChangeFormer [22]	89.97	88.26	91.74	81.76	99.00	89.44	94.61	94.51	94.71	89.77	98.73	93.89	81.10	77.08	85.56	68.21	96.78	79.34
	ELGC-Net [24]	90.59	89.10	92.14	82.80	99.06	90.10	96.47	96.38	96.56	93.18	99.17	96.00	82.23	76.61	88.75	69.83	97.03	80.62
	ConvFormer-CD/48 [26]	91.08	90.09	92.10	83.63	99.11	90.60	96.04	96.21	95.87	92.38	99.09	95.51	85.36	81.80	89.25	74.46	97.48	83.99
	STRobustNet [25]	90.71	89.32	92.15	83.01	99.07	90.22	93.40	92.56	94.25	87.61	98.46	92.52	81.93	81.60	82.26	69.38	96.77	80.15
Mamba-based	ChangeMamba-B [29]	90.09	89.33	90.86	81.01	99.02	89.49	95.09	94.51	95.68	90.64	98.85	94.44	83.81	80.38	87.56	72.14	97.21	82.29
	RSM-CD [30]	91.04	90.24	91.85	83.53	99.11	90.56	96.77	95.86	97.70	93.75	99.17	96.30	85.59	80.88	90.87	74.81	97.56	84.26
	CDMamba [31]	90.56	89.66	91.47	82.74	99.06	90.06	95.34	96.09	94.59	91.09	98.94	94.72	84.63	79.74	90.17	73.36	97.40	83.22
	Ours	91.74	90.76	92.75	84.75	99.17	91.31	97.71	98.11	97.30	95.51	99.46	97.40	86.40	82.10	91.18	76.06	97.68	85.14

5) Overall Accuracy: It measures the proportion of correctly classified pixels among all pixels

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}}. \tag{27}$$

6) Kappa (κ): It evaluates the degree to which the model's predictions surpass random chance, making it especially suitable for imbalanced classification problems such as CD

$$\mathrm{pe} = \frac{(\mathrm{TP} + \mathrm{FN})(\mathrm{TP} + \mathrm{FP}) + (\mathrm{TN} + \mathrm{FN})(\mathrm{TN} + \mathrm{FP})}{(\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN})^2} \tag{28}$$

$$\kappa = \frac{\mathrm{OA} - \mathrm{pe}}{1 - \mathrm{pe}}. \tag{29}$$

In the above formula, TP denotes the number of pixels that truly change and are correctly predicted as change. True negative (TN) denotes the number of pixels that truly remain unchanged and are correctly predicted as unchanged. False positive (FP) denotes the number of pixels that remain unchanged but are incorrectly predicted as change. False negative (FN) denotes the number of pixels that change but are incorrectly predicted as unchanged.

C. Implementation Details

The proposed LGMM-Net is implemented on the PyTorch framework. The encoder (DLGPE) consists of four stages, containing 3, 3, 4, 3 DLGPE blocks, with output channel dimensions of 64, 96, 128, 256. The decoder is composed of five PMM blocks, whose detailed layer configurations are presented in the ablation study (see Section IV-E). Each PMM block adopts a maximum dilation rate of n from deep to shallow layers. All experiments are conducted on a single NVIDIA A40 GPU. To improve robustness and generalization, we apply several data augmentation strategies, including rotation, cropping, Gaussian blur, and color jittering. The model is trained with the AdamW optimizer for 300 epochs, with a weight decay of 0.0001 and default values for β_1 and β_2. The initial learning rate lr_0 is set to 0.001 and decayed following the polynomial (poly) schedule, defined as:

$$\mathrm{lr} = \mathrm{lr}_0 \times \left(1 - \frac{\mathrm{cur_iter}}{\mathrm{total_iter}}\right)^{1.3} \tag{30}$$

where total_iter denotes the total number of training iterations, and cur_iter is the current iteration. On the validation set, the optimal model is selected based on a combination of F1 and loss.

D. Comparative Experiments

To validate the effectiveness of the proposed LGMM-Net, we conducted comparative experiments against eleven representative methods: FC-EF [14], FC-Siam-Diff [14], FC-Siam-Conc [14], BIT [21], ChangeFormer [22], ELGC-Net [24], ConvFormer-CD [26], STRobustNet [25], ChangeMamba [29], RSM-CD [30], and CDMamba [31]. Among them, BIT, ChangeFormer, ELGC-Net, ConvFormer-CD, and STRobustNet are transformer-based approaches. In contrast, ChangeMamba, RSM-CD, and CDMamba are Mamba-based. Although LGMM-Net employs a transformer encoder for feature representation, its Mamba-based decoder constitutes the core innovation of our framework. Therefore, LGMM-Net is classified as a Mamba-based method in Table I. For fairness, all competing models were reproduced using their official open-source implementations within the PyTorch framework. They were executed under the same hardware environment and trained strictly following their default hyperparameter settings.

1) Quantitative Results: Table I reports the performance of the compared methods and the proposed LGMM-Net on the LEVIR-CD, CDD, and GZ-CD datasets across key evaluation metrics. The best and second-best results are highlighted in red and blue, respectively. Overall, LGMM-Net achieves consistently high precision while maintaining superior recall. This demonstrates its ability to better balance the suppression of false changes with the detection of true changes. Notably, on the CDD dataset, although RSM-CD attains the highest precision, its relatively low recall significantly degrades overall performance in terms of F1 and IoU. This suggests that its strategy may be overly aggressive in filtering false changes, thereby sacrificing sensitivity to real ones. In contrast, LGMM-Net delivers more balanced and robust results across all metrics. To further evaluate its effectiveness, we conduct quantitative analysis from two perspectives. First,

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TABLE II
COMPARISON WITH OTHER METHODS ON UAV-CD+ DATASET. THE BEST RESULTS ARE IN RED AND THE NEXT-BEST IN BLUE. ALL VALUES ARE IN %. "/48" DENOTES THE CONVFORMER-CD VARIANT WITH A BASE CHANNEL WIDTH OF 48

Type	Method	UAV-CD+			
		F1	IoU	OA	Kappa
CNN-based	FC-EF	51.30	34.50	92.12	47.75
	FC-Siam-Diff	52.65	35.74	92.20	49.04
	FC-Siam-Conc	51.91	35.05	92.50	48.95
Transformer-based	BIT	61.12	44.01	93.01	57.56
	ChangeFormer	67.60	51.07	93.51	64.06
	ELGC-Net	67.93	51.43	92.33	64.25
	ConvFormer-CD/48	69.19	52.74	93.28	65.47
	STRobustNet	69.06	52.74	93.25	65.56
Mamba-based	ChangeMamba-B	68.32	51.90	92.74	64.24
	RSM-CD	67.31	50.72	92.39	63.61
	CDMamba	69.46	53.20	93.67	66.12
	Ours	70.10	53.96	93.75	66.63

we measure the performance gains of LGMM-Net over the strongest transformer- and Mamba-based baselines using F1 and IoU. Second, we compare its results with the second-best values across all metrics to highlight its balanced yet superior performance.

On the LEVIR-CD dataset, compared with the transformer-based ConvFormer-CD and the Mamba-based RSM-CD, LGMM-Net achieves improvements of 0.66%/1.12% and 0.70%/1.22% in F1 and IoU, respectively. Furthermore, relative to the next-best values of 91.08%, 90.24%, 92.23%, 83.63%, 99.11%, and 90.60% in F1, recall, precision, IoU, OA, and Kappa, LGMM-Net achieves additional gains of 0.66%, 0.52%, 0.52%, 1.12%, 0.06%, and 0.71%, respectively. These results demonstrate that LGMM-Net consistently outperforms competing methods across most evaluation metrics, with particularly notable advantages in F1 and IoU.

On the more challenging CDD dataset, the advantage of LGMM-Net becomes even more pronounced. Compared with the transformer-based ELGC-Net and the Mamba-based RSM-CD, LGMM-Net achieves improvements of 1.24%/2.33% and 0.94%/1.76% in F1/IoU, respectively. Relative to the next-best values of 96.77%, 96.38%, 93.75%, 99.17%, and 96.30% in F1, Recall, IoU, OA, and Kappa, LGMM-Net yields additional gains of 0.94%, 1.73%, 1.76%, 0.29%, and 1.10%. These results demonstrate that LGMM-Net delivers a more balanced and robust performance across multiple evaluation metrics.

Similarly, the experimental results on the GZ-CD dataset further confirm the effectiveness of LGMM-Net. Compared with the transformer-based ConvFormer-CD and the Mamba-based RSM-CD, LGMM-Net achieves improvements of 1.04%/1.60% and 0.81%/1.25% in F1/IoU, respectively. Relative to the next-best values of 85.59%, 81.80%, 90.87%, 74.81%, 97.56%, and 84.26% in F1, recall, precision, IoU, OA, and Kappa, LGMM-Net provides additional gains of 0.81%, 0.30%, 0.31%, 1.25%, 0.12%, and 0.88%, respectively.

As shown in Table II, LGMM-Net demonstrates excellent performance on the UAV-CD+ dataset. Compared with the transformer-based ConvFormer-CD/48 and the Mamba-based CDMamba, LGMM-Net achieves absolute improvements of 0.91%/1.22% and 0.64%/0.76% in F1 and IoU, respectively. Taking the next-best results across all methods, 69.46%, 53.20%, 93.67%, and 66.12% for F1, IoU, OA, and Kappa as reference, LGMM-Net further surpasses them by 0.64%, 0.76%, 0.08%, and 0.51% on the same metrics. Notably, UAV-CD+ is a low-altitude, high-resolution UAV dataset that captures complex urban scenes across different seasons, introducing challenges such as registration errors, scale diversity, and illumination variations. These factors greatly increase the difficulty of detecting small objects and boundary changes. Despite these challenges, LGMM-Net remains robust and achieves a clear performance advantage.

This consistent superiority across architectures and datasets provides strong evidence for the effectiveness of our model design. Notably, although ConvFormer-CD also leverages window-based attention, LGMM-Net shows stronger competitiveness on all four datasets due to its DLGPE module. Furthermore, while RSM-CD adopts a more complex eight-directional scanning strategy, LGMM-Net achieves superior performance with only four directions. This advantage is primarily attributed to the carefully designed PMM module. In summary, the quantitative results clearly validate both the advancement and the effectiveness of the LGMM-Net architecture.

While the quantitative metrics above demonstrate the statistical superiority of LGMM-Net, visual inspections provide deeper insights into its robustness in real-world scenarios. To this end, we next present a qualitative analysis illustrating the generated change maps across different datasets.

2) Qualitative Results: To intuitively evaluate the proposed LGMM-Net and further verify its effectiveness, we conducted qualitative comparisons on three public datasets: LEVIR-CD, CDD, and GZ-CD. Figs. 6–8 present the visual results of LGMM-Net against several SOTA methods across six representative samples. In all visualizations, white denotes TPs, black denotes TNs, red indicates FPs (i.e., pseudo-changes), and cyan indicates FNs (i.e., missed-detection regions).

Fig. 6 shows qualitative comparison results on the LEVIR-CD dataset. The six selected samples cover diverse building-change scenarios, ranging from dense to sparse regions and from large-scale to small-scale changes. In dense building-change scenarios, shown in samples (a) and (b), LGMM-Net demonstrates strong robustness. It effectively suppresses pseudo-changes caused by nontarget variations such as roads and shadows, which significantly reduces false alarms. In the sparse-change scenarios, shown in sample (c), most competing methods either show low sensitivity, leading to large missed-detection regions (cyan regions), or excessive sensitivity, introducing FPs (red regions). By contrast, LGMM-Net maintains high recall while introducing almost no pseudo-changes. In the small-scale building-change scenarios, represented by sample (d), LGMM-Net accurately captures fine-grained change boundaries. For large-scale changes scenarios, illustrated in samples (e) and (f), LGMM-Net achieves superior detection completeness. It delineates change regions precisely and avoids the missed-detection regions commonly observed in other models.

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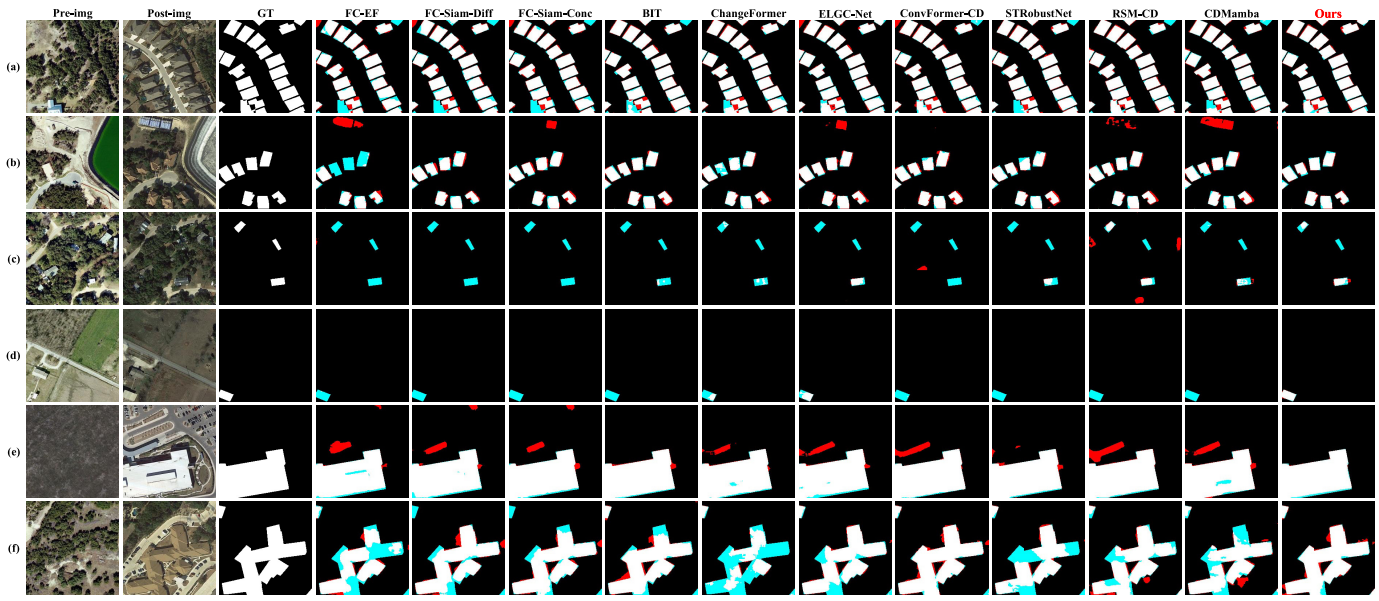


Fig. 6. Qualitative comparison results on the LEVIR-CD dataset. (a)–(f) Prediction maps of six representative image pairs across all methods. In the visualization, white denotes true changes regions (TP), black denotes unchanged regions (TN), red denotes pseudo-changes regions (FP), and cyan denotes missed-detection regions (FN).

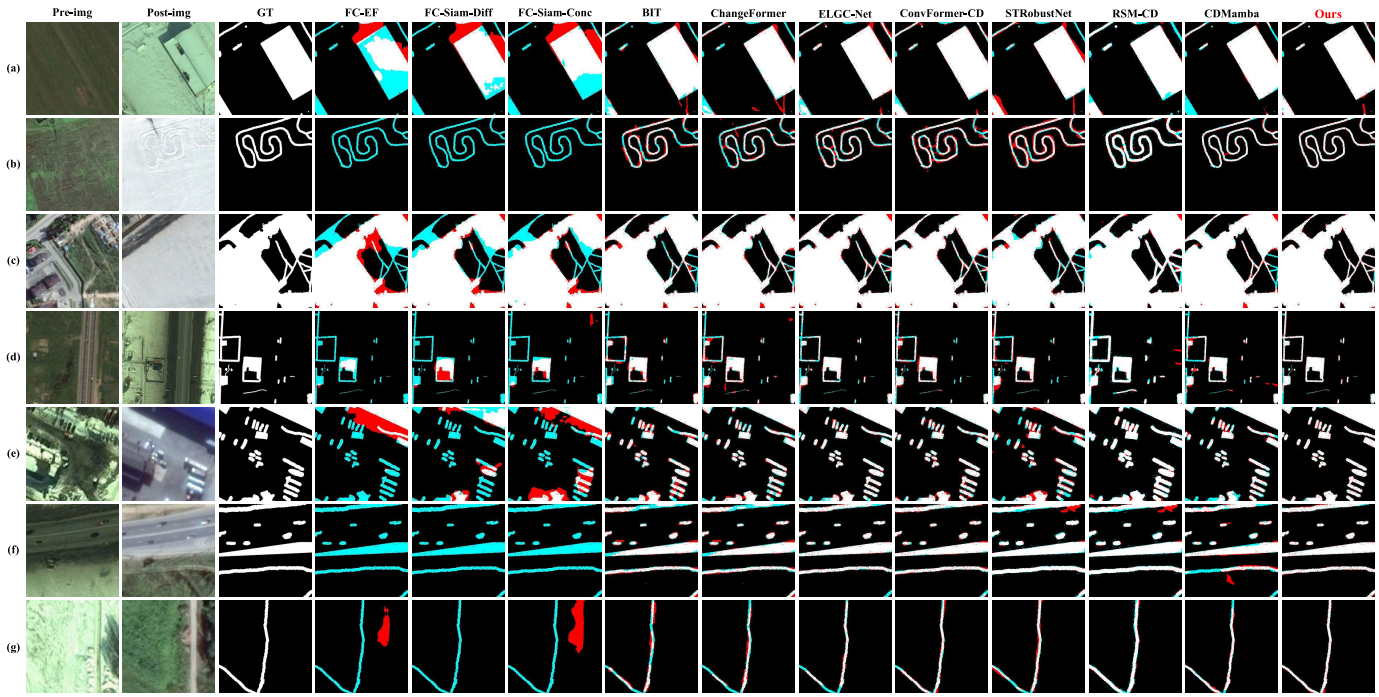


Fig. 7. Qualitative comparison results on the CDD dataset. (a)–(g) Prediction maps of six representative image pairs across all methods. In the visualization, white denotes true changes regions (TP), black denotes unchanged regions (TN), red denotes pseudo-changes regions (FP), and cyan denotes missed-detection regions (FN).

Fig. 7 shows qualitative results on the CDD dataset. This dataset contains both real object changes, such as buildings, roads, and vehicles, as well as strong seasonal disturbances. While competing methods attempt to detect change regions, seasonal interference often causes numerous FPs and FNs. In large-scale building-change scenarios shown in sample (a), LGMM-Net generates detection maps with complete contours. In contrast, other methods are heavily affected by seasonal background variations, which result in substantial misdetections. In fine-grained road-change scenarios illustrated in samples (b) and (c), LGMM-Net accurately identifies linear feature changes. Competing models, however, struggle to balance the suppression of pseudo-changes with the need to ensure detection completeness. In multiscale mixed scenarios illustrated in samples (d) and (e), which include both vehicles and buildings, LGMM-Net detects large-scale building

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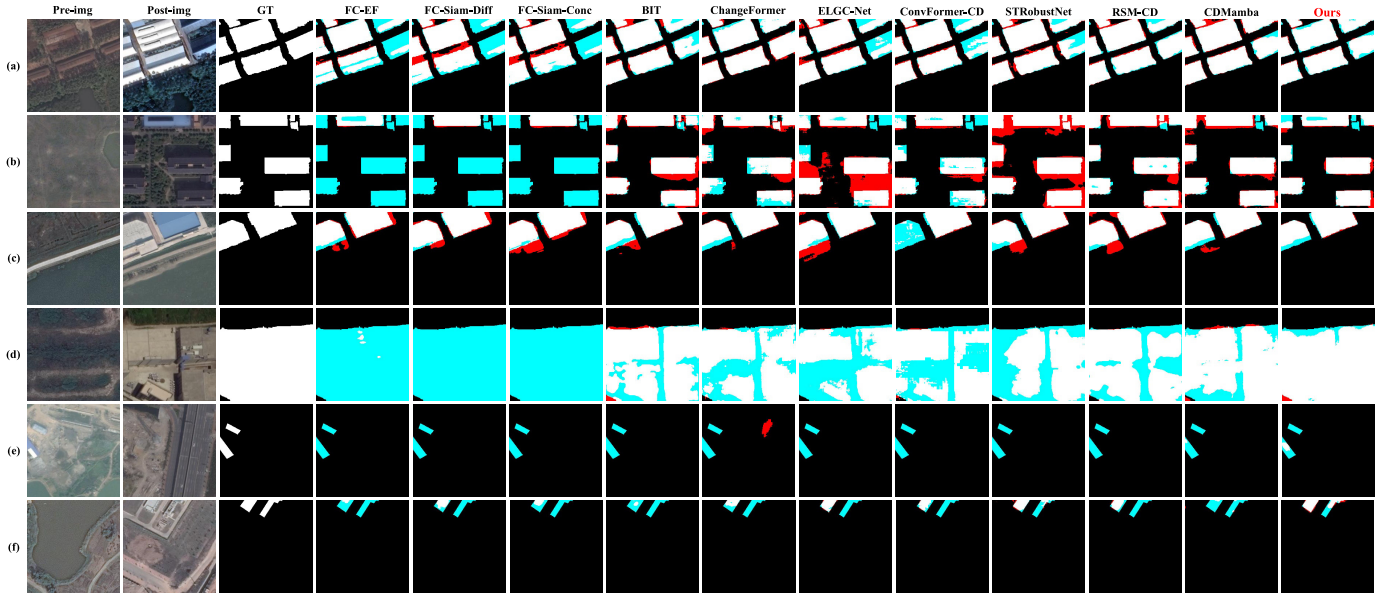


Fig. 8. Qualitative comparison results on the GZ-CD dataset. (a)–(f) Prediction maps of six representative image pairs across all methods. In the visualization, white denotes true changes regions (TP), black denotes unchanged regions (TN), red denotes pseudo-changes regions (FP), and cyan denotes missed-detection regions (FN).

changes while simultaneously capturing small-scale vehicle changes. In the complex case shown in sample (f), most competing methods recognize only isolated small targets such as vehicles but fail to capture broader road changes. By contrast, LGMM-Net achieves more comprehensive detection of both vehicles and roads, demonstrating strong adaptability to multiscale targets. In the extreme pseudo-change scenario shown in sample (g), strong seasonal and illumination variations cause obvious spectral differences without real land-cover changes. Competing methods produce numerous FPs and FNs, while LGMM-Net effectively suppresses these pseudo-change responses and preserves true change boundaries, showing high robustness to seasonal disturbances.

Fig. 8 shows qualitative results on the GZ-CD dataset. In dense and irregular building-change scenarios such as samples (a)–(d), competing methods struggle to delineate boundaries and capture fine edges, leading to many FNs and FPs. By contrast, LGMM-Net produces sharper boundaries and more complete interiors, which markedly reduces both error types. In the small-object case illustrated in sample e, all methods exhibit large missed regions. However, LGMM-Net is the only model that successfully recovers the changes. In another small-object case shown in sample f, competing methods often miss changes or generate false alarms. LGMM-Net, in contrast, preserves true change regions while suppressing spurious responses, thus producing cleaner maps.

In summary, the qualitative comparisons clearly demonstrate the superiority of LGMM-Net across multiple scales and diverse scenarios. Whether in large-scale, small-scale, dense, or sparse cases, LGMM-Net consistently achieves leading performance. On the CDD dataset, where seasonal variations present severe challenges, LGMM-Net shows strong robustness. It effectively separates true changes from pseudo-changes and accurately extracts change information. These visual results provide intuitive evidence of the advantages of LGMM-Net in detection accuracy, completeness, and adaptability to complex environments.

TABLE III

MODEL EFFICIENCY COMPARISON. PARAMS DENOTE THE NUMBER OF PARAMETERS (M), AND FLOPS INDICATE THE COMPUTATIONAL COST (GFLOPS). F1 (%) AND KAPPA (%) ARE THE PERFORMANCE SCORES OBTAINED ON THE LEVIR-CD DATASET. INFER REPRESENTS THE AVERAGE INFERENCE TIME PER IMAGE (MS) AND FPS INDICATES FRAMES PER SECOND, BOTH MEASURED USING 256 × 256 INPUT PATCHES ON A SINGLE NVIDIA A40 GPU UNDER IDENTICAL TESTING CONDITIONS. "/48" DENOTES THE CONVFORMER-CD VARIANT WITH A BASE CHANNEL WIDTH OF 48

Model	Params	FLOPs	F1	Kappa	Infer	FPS
FC-EF	1.35	3.58	86.31	85.58	11.47	87.16
FC-Siam-Diff	1.35	4.73	88.42	87.81	11.54	86.65
FC-Siam-Conc	1.55	5.34	89.29	88.72	11.48	87.14
BIT	3.50	10.63	90.25	89.74	29.79	33.56
ChangeFormer	41.03	202.79	89.97	89.44	56.77	17.61
ELGC-Net	10.57	188.21	90.59	90.10	54.35	18.40
ConvFormer-CD/48	49.33	6.72	91.08	90.60	63.47	15.76
STRobustNet	13.73	20.99	90.71	90.22	72.37	13.82
ChangeMamba-B	59.41	31.91	90.09	89.49	91.31	10.73
RSM-CD	51.95	18.35	91.04	90.56	213.00	4.69
CDMamba	11.91	49.26	90.56	90.06	252.28	3.96
Ours	15.18	15.39	**91.74**	**91.31**	160.02	6.25

3) Model Efficiency: To further evaluate the computational efficiency and deployment potential of LGMM-Net, Table III provides a comprehensive comparison with competing methods in terms of parameter count (Params), floating-point operations (FLOPs), and key performance metrics (F1 and Kappa). Our analysis highlights the tradeoff advantages of LGMM-Net across different efficiency dimensions. Compared with CDMamba, which has the fewest parameters (11.91 M), LGMM-Net reduces FLOPs by 68.7% (from 49.26 GFLOPs to 15.39 GFLOPs) with only a slight increase in parameters.

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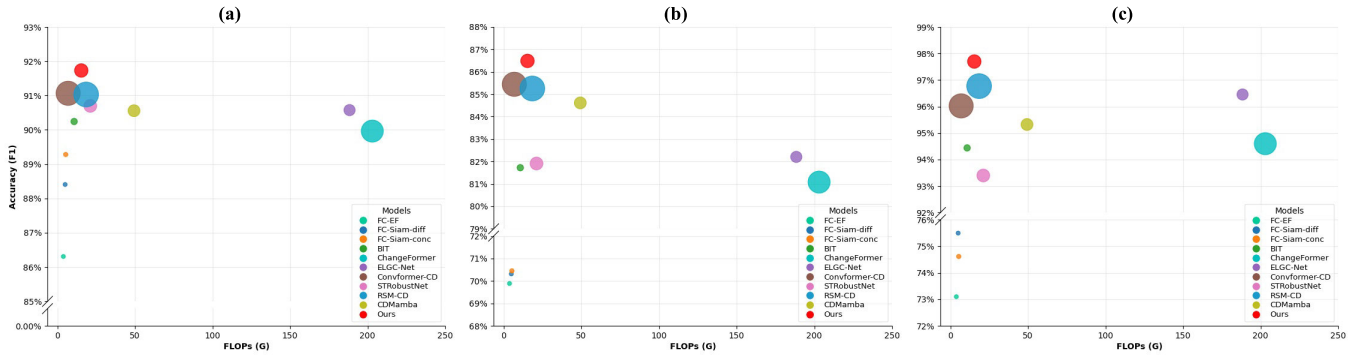


Fig. 9. Bubble chart comparison of different methods on three datasets: (a) LEVIR-CD, (b) GZ-CD, and (c) CDD. The vertical axis represents the F1, while the horizontal axis indicates computational complexity (GFLOPs). The bubble size is proportional to the number of model parameters.

TABLE IV

MACRO-LEVEL ABLATION STUDY OF LGMM-NET ON THE LEVIR-CD AND CDD DATASETS. THIS STUDY FOCUSES ON EVALUATING THE EFFECTIVENESS OF ENTIRE MODULES. IN THE TABLE, ELGCA DENOTES THE ENCODER FROM ELGC-NET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD, AND ALL NUMBERS ARE PERCENTAGES (%)

ID	Variant					Complexity		LEVIR-CD						CDD					
	Baseline	DLGPE	PMM	ELGCA	ResNet18	Params (M)	FLOPs (G)	F1	Rec.	Pre.	IoU	OA	Kappa	F1	Rec.	Pre.	IoU	OA	Kappa
#01	✓					12.38	103.73	90.54	88.90	92.24	82.71	99.05	90.04	96.65	96.30	97.01	93.53	99.18	96.21
#02	✓	✓				14.25	105.06	91.17	89.99	92.38	83.77	99.11	90.70	97.36	97.63	97.09	94.85	99.37	97.00
#03	✓		✓			13.34	14.12	91.05	89.18	92.99	83.56	99.11	90.58	97.39	97.68	97.10	94.91	99.38	97.04
#04	✓		✓	✓		13.59	13.51	91.19	89.87	92.55	83.81	99.12	90.72	97.51	97.79	97.23	95.14	99.41	97.18
#05	✓		✓		✓	12.59	13.51	91.12	89.41	92.88	83.68	99.11	90.65	97.13	97.48	96.79	94.43	99.32	96.75
#06	✓	✓	✓			15.18	15.39	**91.74**	**90.76**	**92.75**	**84.75**	**99.17**	**91.31**	**97.71**	**98.11**	**97.30**	**95.51**	**99.46**	**97.40**

At the same time, it improves F1 and Kappa by 1.18% and 1.25%, respectively. Compared with ConvFormer-CD, which has the lowest FLOPs (6.72 GFLOPs), LGMM-Net reduces the parameter count by 69.2% (from 49.33 to 15.18 M) while incurring only a moderate increase in FLOPs. It further achieves performance gains of 0.66% in F1 and 0.71% in Kappa. This advantage is also illustrated in Fig. 9, where LGMM-Net (red marker) consistently occupies the upper-left "high-efficiency" region among all methods. Also, LGMM-Net achieves an inference speed of 160.02 ms/image, corresponding to 6.25 FPS, when evaluated on 256 × 256 input patches. Although its efficiency is lower than that of purely convolution-based and transformer-based architectures, it still exhibits competitive inference performance within the Mamba-based family, reflecting the efficiency benefits of our design. Overall, LGMM-Net does not aim for extreme optimization on a single efficiency indicator. Instead, it maintains a balanced tradeoff among parameter size, FLOPs, and accuracy, achieving state-of-the-art (SOTA) performance while providing reasonable computational efficiency without sacrificing representational capacity.

E. Ablation Studies

To systematically validate the effectiveness of each innovative component in LGMM-Net and to demonstrate the rationality of its internal design, we conducted a series of ablation studies on the LEVIR-CD and CDD datasets. The baseline model was built on a simplified version of the SOTA ELGC-Net. Specifically, we replaced its core ELGCA encoder with a standard self-attention network while retaining the decoder structure. The fusion module between the encoder and decoder remained the same as in LGMM-Net, consisting of two cascaded MV2 blocks. All subsequent variant experiments were performed on this baseline. All models were trained under identical conditions, including data augmentation (random rotation and Gaussian blur), optimizer settings, and 300 training epochs. The detailed quantitative results are reported in Table IV.

1) Effectiveness Analysis of DLGPE and PMM: Aiming to validate the effectiveness of the proposed DLGPE and PMM, we conduct a series of ablation experiments. As shown in row #02 of Table IV, replacing the baseline encoder with our DLGPE yields a significant performance improvement. Specifically, with only a slight increase in parameters and FLOPs, the model achieves gains of 0.63%/1.06%/0.66% in F1/IoU/Kappa on the LEVIR-CD dataset, and 0.71%/1.32%/0.79% on the CDD dataset. These results demonstrate that DLGPE delivers substantial performance gains with minimal additional computational cost. Similarly, as shown in row #03 of Table IV, replacing the baseline decoder with our PMM yields clear improvements. Notably, this gain is achieved while substantially reducing FLOPs. On LEVIR-CD and CDD, the model achieves improvements of 0.51%/0.85%/0.54% and 0.74%/1.38%/0.83% in F1/IoU/Kappa, respectively. These results validate both the efficiency and the effectiveness of the PMM decoder. Finally, as shown in row #06 of Table IV, combining DLGPE and PMM to form the complete LGMM-Net further boosts performance. The overall model not only surpasses the baseline but also outperforms any single-module variant. Compared with the baseline, LGMM-Net reduces FLOPs by 85% (from 103.73 to 15.39 G) while

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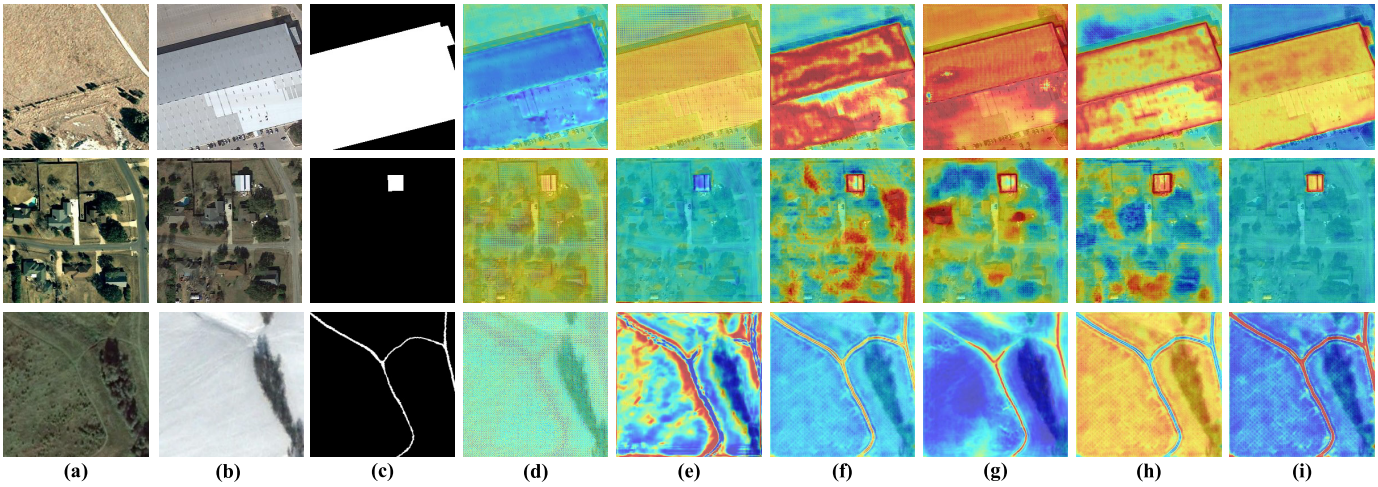


Fig. 10. Visual comparison of pixel-level feature responses for the macro-level ablation study. Each row corresponds to a representative scenario: large buildings, small objects, and linear roads. (a) Prechange image, (b) postchange image, and (c) GT. (d)–(i) Pixel-wise response heatmaps of variants #01–#06 in Table IV. Warmer colors indicate higher responses, while cooler colors indicate lower responses.

achieving gains of 1.20%/2.04%/1.27% on LEVIR-CD and 1.06%/1.98%/1.19% on CDD in F1/IoU/Kappa. These findings clearly demonstrate the positive synergy between DLGPE and PMM, which together ensure both efficiency and superior performance.

Furthermore, to intuitively illustrate the effect of each component, Fig. 10 presents the pixel-level activation heatmaps for four representative variants: (d) the baseline, (e) baseline + DLGPE, (f) baseline + PMM, and (i) the complete LGMM-Net. In large buildings, small objects, and linear roads scenarios, the baseline model fails to focus precisely on the actual changed regions, showing dispersed or background-biased activations. After integrating DLGPE, the model becomes more sensitive to structural and spatial variations, particularly in large-scale targets and linear road areas, demonstrating its capability to enhance global–local contextual modeling. In contrast, the PMM-enhanced variant highlights detailed and fine-grained changes, effectively improving attention to small objects and subtle edge regions. Finally, the complete LGMM-Net integrates the strengths of both modules, accurately localizing diverse change patterns across all scenarios, with concentrated and well-aligned activations around the true change areas. These qualitative results further confirm that DLGPE and PMM are complementary, jointly enabling LGMM-Net to achieve comprehensive and consistent change awareness.

2) Comparison of DLGPE With Different Encoders: To further highlight the superiority of DLGPE as an encoder, we compared it with the original encoder of ELGC-Net (ELGCA) and the classical CNN backbone ResNet18. In this experiment, the PMM decoder was kept fixed, while only the encoder was replaced. As shown in row #04 of Table IV, replacing DLGPE with ELGCA led to decreases of 0.55%/0.94%/0.58% in F1/IoU/Kappa on LEVIR-CD and 0.17%/0.32%/0.19% on CDD, compared with the full LGMM-Net (row #06 of Table IV). Similarly, as shown in row #05 of Table IV, replacing the encoder with ResNet18 resulted in a more pronounced performance drop, with reductions of 0.63%/1.07%/0.66% on LEVIR-CD and 0.54%/1.03%/0.62% on CDD. These comparative results clearly demonstrate that the proposed DLGPE exhibits superior feature extraction capability over both ELGCA and conventional CNN.

To further provide qualitative evidence for the superiority of DLGPE as an encoder, Fig. 10(g)–(i) visualizes the pixel-level activation maps of different encoder variants: (g) baseline + DLGPE + PMM, (h) baseline + ResNet18 + PMM, and (i) the complete LGMM-Net. When the proposed DLGPE is replaced by ELGCA, the model primarily attends to small objects and linear structures while failing to effectively capture large-scale building changes. In contrast, replacing DLGPE with ResNet18 enables partial sensitivity to large buildings and small targets. However, the activation responses appear weak and inconsistent, and the model shows almost no focus on linear road regions. By comparison, the complete LGMM-Net achieves strong and well-localized responses across all three scenarios—large buildings, small objects, and linear roads—demonstrating comprehensive and balanced change perception. These visual results align with the quantitative findings in Table IV, further verifying the superior feature representation capability of DLGPE.

3) Effectiveness of the Internal Design of DLGPE: To further validate the effectiveness and rationality of the DLGPE, we conducted a detailed ablation study on its key components and configurations using the LEVIR-CD and GZ-CD datasets. The quantitative results are reported in Table V. The core of DLGPE lies in its parallel three-branch design: the multiconv branch, the fw_attn branch, and the sw_attn branch. To demonstrate the indispensability of these branches, we performed ablation experiments on different combinations (rows #01–#07 of Table V). We first evaluated the complete three-branch model (row #07 of Table V), achieving F1/IoU scores of 91.74%/84.75% on LEVIR-CD and 86.40%/76.06% on GZ-CD, respectively. When removing branches and retaining only a single one (rows #01–#03 of Table V), performance dropped significantly compared with the full design. The F1/IoU/Kappa scores decreased by up to 0.36%/0.63%/0.39%

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TABLE V

MICRO-LEVEL ABLATION STUDY OF THE DLGPE MODULE ON THE LEVIR-CD AND GZ-CD DATASETS, TARGETING THE INTERNAL DESIGN OF EACH COMPONENT FOR FINE-GRAINED ANALYSIS. IN THE TABLE, DWCONV DENOTES A DILATED 3×3 CONVOLUTION WITH DILATION = 2. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD, AND ALL NUMBERS ARE REPORTED AS PERCENTAGES (%)

ID	Variant	LEVIR-CD			GZ-CD		
		F1	IoU	Kappa	F1	IoU	Kappa
(a) Branch Combinations in DLGPE							
#01	multi-conv	91.38	84.12	90.92	85.41	74.53	84.08
#02	fw_attn	91.44	84.23	90.98	84.60	73.31	83.15
#03	sw_attn	91.53	84.38	91.08	85.21	74.23	83.82
#04	fw_attn + sw_attn	91.55	84.42	91.11	86.08	75.56	84.83
#05	multi-conv + sw_attn	91.38	84.12	90.92	86.20	75.74	84.98
#06	multi-conv + fw_attn	91.52	84.37	91.08	85.66	74.92	84.34
#07	multi-conv + fw_attn + sw_attn	**91.74**	**84.75**	**91.31**	**86.40**	**76.06**	**85.14**
(b) Hard Channel Split							
#08	hard split	91.41	84.19	90.96	84.18	72.68	82.71
(c) Multi-kernel Convolution Configurations							
#09	kernel setting: {2,2,2,2}	91.26	83.93	90.88	86.20	75.74	84.90
#10	kernel setting: {3,3,3,3}	91.45	84.25	91.00	86.05	75.52	84.79
#11	kernel setting: {4,3,3,2}	**91.74**	**84.75**	**91.31**	**86.40**	**76.06**	**85.14**
#12	kernel setting: {4,4,4,4}	91.42	84.20	90.97	86.21	75.76	84.91
(d) QKV Generation Strategies for Attention							
#13	fw_attn = Linear, sw_attn = Linear	91.39	84.14	90.93	86.00	75.44	84.67
#14	fw_attn = PW, sw_attn = PW	91.38	84.13	90.92	86.02	75.47	84.72
#15	fw_attn = DWConv, sw_attn = PW	91.47	84.28	91.02	86.13	75.64	84.89
#16	fw_attn = PW, sw_attn = DWConv	**91.74**	**84.75**	**91.31**	**86.40**	**76.06**	**85.14**
#17	fw_attn = DWConv, sw_attn = DWConv	91.58	84.47	91.13	86.18	75.72	84.95

on the LEVIR-CD dataset and 1.80%/2.75%/1.99% on the GZ-CD dataset.

We then evaluated two-branch combinations (rows #04–#06 of Table V). Although these achieved better results than single branches, their performance remained inferior to the complete three-branch structure. The best two-branch case (row #04 of Table V) still lagged behind the complete model by 0.19%/0.33%/0.20% on LEVIR-CD and 0.32%/0.50%/0.31% on GZ-CD in F1/IoU/Kappa.

To further quantify the relative contribution of each DLGPE branch, we compute branch-level Shapley values using all 2^3 coalitions implied by Table V. Treating each branch as a player, the Shapley value of branch i is defined as

$$\phi_i = \sum_{S \subseteq \mathcal{N} \setminus \{i\}} \frac{|S|!\,(|\mathcal{N}| - |S| - 1)!}{|\mathcal{N}|!} \left(m(S \cup \{i\}) - m(S)\right). \quad (31)$$

Intuitively, ϕ_i measures the expected F1 gain contributed by adding branch i to a randomly chosen subset of the other branches. Here, $\mathcal{N} = \{\text{sw_attn}, \text{multi-conv}, \text{fw_attn}\}$, S is a subset (coalition) that excludes i (i.e., $S \subseteq \mathcal{N} \setminus \{i\}$), and $S \cup \{i\}$ denotes adding branch i to S. $m(S)$ denotes the evaluation score (e.g., test F1) obtained by enabling exactly the branches in S under the same fixed backbone, and $m(\varnothing)$ denotes the empty-coalition score.

Shapley computation requires $m(\varnothing)$. Since Table V reports branch combinations on top of a fixed baseline+PMM backbone and only toggles DLGPE branches, we set $m(\varnothing)$ to the performance of the baseline+PMM model (Table IV, #03). All coalition models $m(S)$ are obtained by inserting DLGPE into this baseline+PMM backbone and enabling only the branches in S, ensuring that all scores differ only in which DLGPE branches are active. The resulting attributions are $\phi_{\text{fw}} = 2.77 \times 10^{-3}$ (40.10%), $\phi_{\text{sw}} = 2.52 \times 10^{-3}$ (36.47%), and $\phi_{\text{mc}} = 1.62 \times 10^{-3}$ (23.43%), where percentages are normalized by $\sum_{j \in \mathcal{N}} \phi_j$. This analysis complements the ablation study by averaging marginal contributions over all coalitions and further confirms that the three branches provide complementary gains.

To better interpret the ablation results, we visualize in Fig. 11 the relative F1 drops caused by different branch removals, which reveals dataset-specific drop patterns. On the LEVIR-CD dataset, removing the fw_attn branch alone or jointly removing both fw_attn and sw_attn branches causes the most significant drop in F1. This pattern aligns with the fact that most changes in LEVIR-CD occur along linear man-made structures, as illustrated in Fig. 6(a) and (b), such as roads and building edges, where the fw_attn and sw_attn branches are crucial for capturing fine-grained spatial transitions and local context. In contrast, on the GZ-CD dataset, the performance declines most sharply when the sw_attn branch

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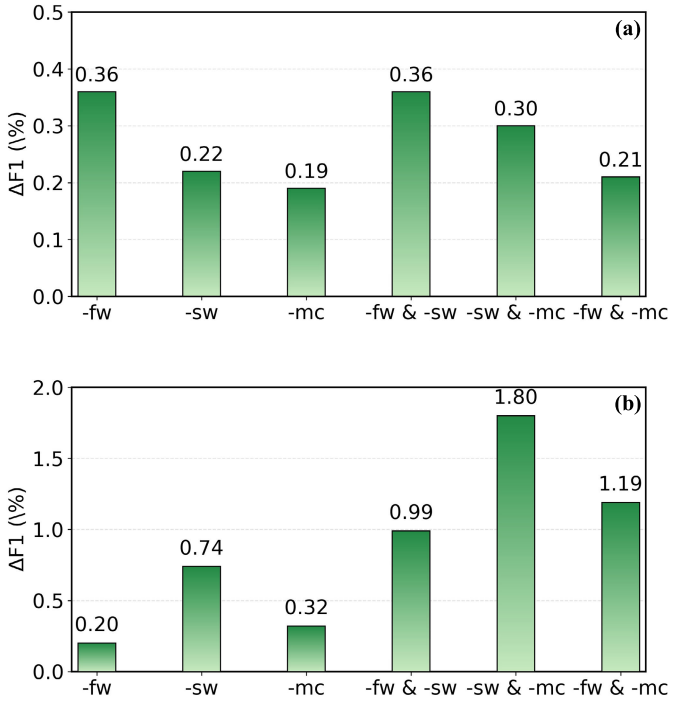


Fig. 11. Performance comparison of different branch combinations in DLGPE. Bars indicate ΔF1 (%). (a) Results on LEVIR-CD; (b) results on GZ-CD. fw, sw, and mc denote the simplified labels of fw_attn, sw_attn, and multiconv, respectively. "-" denotes the removal operation. -fw, -sw, and -mc denote the removal of the corresponding branch from the full DLGPE model. -fw and -sw keep only the multiconv branch, -sw and -mc keep only the fw_attn branch, and -fw and -mc keep only the sw_attn branch.

is removed, particularly in the sw_attn and multiconv combination. As shown in Fig. 8(a) and (b), most changes in GZ-CD exhibit irregular textures and large-scale intensity variations. These patterns rely heavily on the sw_attn branch to model broader semantic dependencies, while the multiconv branch complements it through enhanced multiscale representation. This dataset-specific behavior further confirms that the three branches contribute distinct yet complementary features for different types of change. These results clearly indicate that the three branches capture complementary feature information, and their synergy is crucial for optimal performance, making the full three-branch design necessary.

In Section III-B, we introduced a soft-split mechanism in DLGPE to replace the traditional hard split. To assess its effectiveness, we compared it with hard split (row #08 of Table V). The model's F1/IoU/Kappa decreased by 0.33%/0.56%/0.35% on the LEVIR-CD and 2.22%/3.38%/2.43% on the GZ-CD relative to the soft-split model (row #07 of Table V). This result confirms that fixed channel-allocation hinders adaptive feature fusion. Hard-splitting forces each branch to process a static portion of features regardless of scene characteristics, which limits the encoder's flexibility and may amplify pseudo-changes in regions dominated by illumination or seasonal variations. In contrast, our soft-splitting mechanism adaptively balances local and global cues, resulting in more stable optimization and higher generalization across datasets. This analysis quantitatively supports our claim that the soft-splitting design fundamentally resolves the core limitation of ELGC-Net's hard-splitting encoder.

To further interpret this adaptive behavior of DLGPE, we visualize the spatial distributions of the learnable weighting factors α_1, α_2, and α_3 across encoder stages in Fig. 12. The visualization provides intuitive evidence of how the soft-splitting mechanism dynamically adjusts feature routing among the three branches according to scene characteristics. In regions with large-scale changes, α_1 and α_3 show higher activation, indicating that the network emphasizes the sw_attn and fw_attn branches to capture long-range dependencies and boundary transitions, while α_2 remains stable to preserve contextual consistency through the multiconv branch. In contrast, for small and localized changes, α_1 and α_2 dominate, suggesting that the model relies more on the sw_attn and multiconv branches for contextual aggregation and multiscale semantics rather than fine-grained attention. These results confirm that the dynamic soft-splitting mechanism adaptively balances local and global representations according to the spatial scale of changes.

We further investigated critical design details within DLGPE. To identify the optimal kernel configuration for the multikernel branch, we tested four classical settings (rows #09–#12 of Table V). The configuration {4, 3, 3, 2} (row #11 of Table V) achieved the best performance. In contrast, both larger and smaller kernel settings degraded performance, validating the rationality of the chosen parameters.

As discussed in Section III-B, the fw_attn and sw_attn branches emphasize local and global modeling, respectively. They should therefore adopt different Q, K, and V generation strategies: 1×1 convolution for fw_attn and 3×3 dilated convolution ($d = 2$) for sw_attn (row #16 of Table V). To verify this, we conducted ablation experiments (rows #13–#17 of Table V). The results show that unifying both branches with the same generation strategy-Linear, PW conv (row #14 of Table V), 3×3 dilated conv ($d = 2$) (row #17 of Table V)-all performances decline. Likewise, swapping the generation methods (row #15 of Table V) also degrades accuracy. The F1/IoU/Kappa scores drop by up to 0.36%/0.62%/0.39% on the LEVIR-CD and 0.40%/0.62%/0.47% on the GZ-CD. These findings strongly support our design choice: adopting distinct Q, K, and V generation strategies tailored to different attention types enables the model to achieve optimal performance.

4) Effectiveness of the Internal Design of PMM: To systematically evaluate the rationality and effectiveness of the internal design and configurations of the PMM module, we conducted a series of detailed ablation studies on the LEVIR-CD and GZ-CD datasets. The quantitative results are reported in Table VI.

To validate the effectiveness of the dynamic residual mechanism introduced in Section III-C, we removed the dynamic adjustment controlled by parameters $W_0 - W_3$ and replaced it with a standard residual connection. As shown in row #02 of Table VI, this variant led to decreases of 0.32%/0.56%/0.35% in F1/IoU/Kappa on the LEVIR-CD and 0.67%/1.04%/0.74% on the GZ-CD compared to the full LGMM-Net model (row #01 of Table VI). These results demonstrate that the

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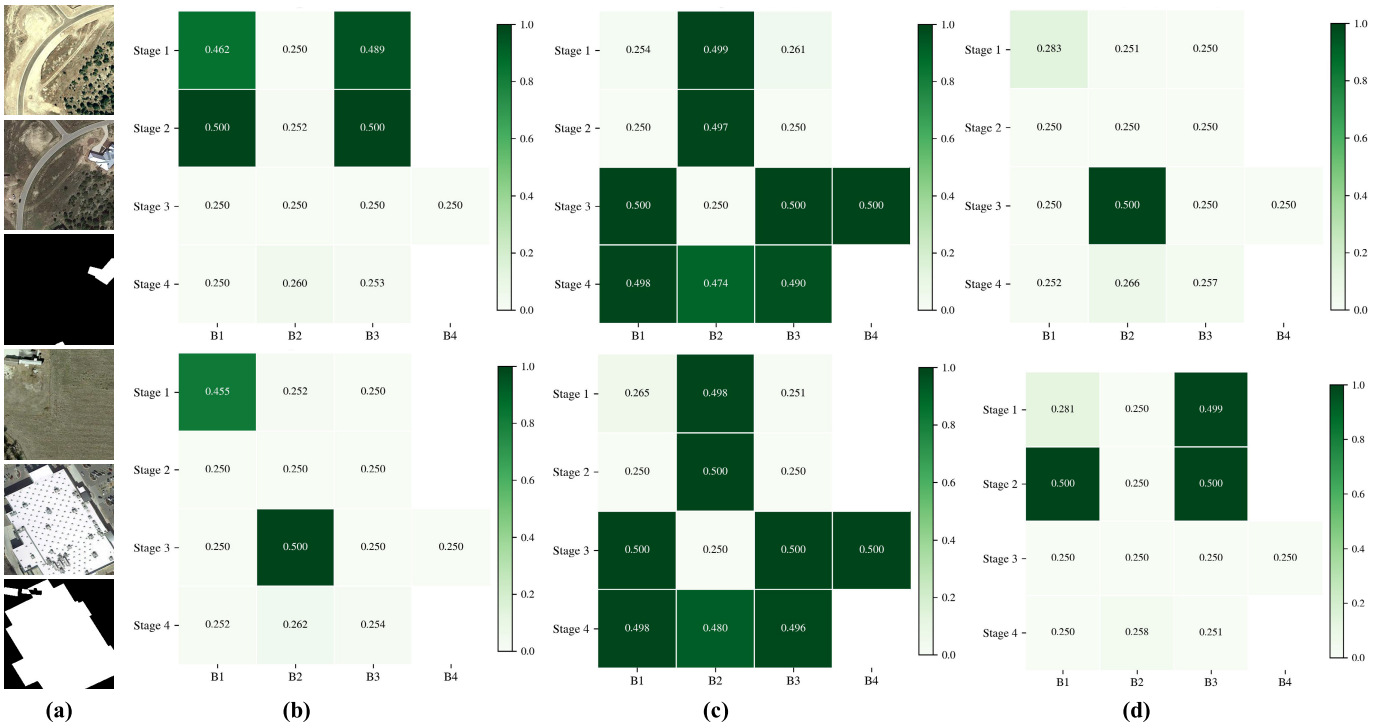


Fig. 12. Visualization of the learnable channel-allocation coefficients α_1, α_2, and α_3 in DLGPE across encoder stages on the LEVIR-CD dataset. (a) Input bi-temporal images (T1/T2) and the corresponding GT change masks. (b)–(d) Heatmaps of α_1, α_2, and α_3, which control the channel-wise soft splitting for the sw-attn, multiconv, and fw-attn branches, respectively. Rows denote the encoder stages (Stages 1–4), and columns denote the stacked DLGPE blocks within each stage (B1–B4). Darker colors mean the model assigns a higher weight to that branch. The top group shows small and localized changes, while the bottom group shows a scene with large-scale changes, highlighting how the branch allocation varies with scene characteristics.

dynamic residual mechanism effectively integrates contextual semantics and contributes to performance improvement.

We also emphasized in Section III-C the importance of constructing directional convolutions via mask-initialized kernels. To verify this design, we removed the mask initialization mechanism (row #03 of Table VI). The results show performance drops of 0.29%/0.51%/0.32% in F1/IoU/Kappa on the LEVIR-CD and 0.71%/1.10%/0.81% on the GZ-CD, confirming that mask initialization plays a critical role in learning effective direction-aware features.

To investigate the effectiveness of different scanning mechanisms for contextual aggregation, we conducted the following ablation experiments. First, we reduced the proposed four-directional MGD-SS2D scanning with the mainstream bidirectional scanning (row #04 of Table VI). In this case, F1/IoU/Kappa dropped by 0.21%/0.36%/0.22% on LEVIR-CD and 0.35%/0.54%/0.38% on GZ-CD. This performance degradation clearly demonstrates that the four-directional scanning captures spatial dependencies more comprehensively than conventional bidirectional scanning. Furthermore, replacing the MGD-SS2D scanning mechanism with an eight-directional scanning mechanism in RSM-CD (row #05 of Table VI). This variant performance declined by 0.43%/0.73%/0.37% on LEVIR-CD and 0.83%/1.28%/0.86% on GZ-CD, showing that excessive directional decomposition introduces redundancy and weakens global context modeling. Finally, we evaluated how rotation-based reverse-scan simulation affects performance under unidirectional scanning. (rows #06–#09 of Table VI). The results show that the 180° rotation (row #08) achieved the best performance, whereas 0°, 90°, and 270° rotations yielded lower performance, with maximum drops of 0.25%/0.43%/0.27% on LEVIR-CD and 1.82%/2.78%/2.05% on GZ-CD in F1/IoU/Kappa. We attribute this phenomenon to the disruption of feature correspondence during sequential modeling. The MGD-SS2D mechanism relies on directional masked kernels designed to encode specific anisotropic priors, such as horizontal continuity. A 180° rotation effectively acts as a spatial flip that inverts the sequence order while preserving the row-major or column-major spatial topology, ensuring that the masked kernel remains aligned with the structural continuity of ground objects. In contrast, 90° or 270° rotations transpose the spatial dimensions, converting row-aligned features into column-aligned ones. This misalignment causes the direction-specific kernel to scan across, instead of along, the continuous structures, thereby disrupting the long-range dependency chain and degrading the representation of object boundaries.

To determine the optimal distribution of PMM in each PMM block across decoder stages, we evaluated four common configurations (rows #10–#13 of Table VI). The results clearly indicate that the configuration {2,3,2,1,1} (row #12 of Table VI) achieves the best performance. Both increasing and decreasing the number of PMM modules led to performance degradation, with maximum drops of 0.32%/0.56%/0.35% on LEVIR-CD and 0.39%/0.60%/0.43% on GZ-CD in F1/IoU/Kappa. This demonstrates that our chosen configuration strikes a favorable balance between

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TABLE VI

MICRO-LEVEL ABLATION STUDY OF THE PMM MODULE ON THE LEVIR-CD AND GZ-CD DATASETS, TARGETING THE INTERNAL DESIGN OF EACH COMPONENT. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD, AND ALL NUMBERS ARE REPORTED AS PERCENTAGES (%)

ID	Variant	LEVIR-CD			GZ-CD		
		F1	IoU	Kappa	F1	IoU	Kappa
#01	full PMM	**91.74**	**84.75**	**91.31**	**86.40**	**76.06**	**85.14**
(a) Dynamic Residual Mechanism							
#02	w/o dynamic residual (W_0–W_3 removed)	91.42	84.19	90.96	85.73	75.02	84.40
(b) Mask-initialized Directional Convolutions							
#03	w/o mask initialization	91.45	84.24	90.99	85.69	74.96	84.33
(c) Scanning Strategies							
#04	bidirectional scanning	91.53	84.39	91.09	86.05	75.52	84.76
#05	8 bidirectional scanning	91.31	84.02	90.94	85.57	74.78	84.28
(d) Rotation-Based Reverse-Scan Simulation							
#06	rotation = 0°	91.50	84.33	91.05	85.36	74.45	84.04
#07	rotation = 90°	91.49	84.32	91.05	84.58	73.28	83.09
#08	rotation = 180°	**91.74**	**84.75**	**91.31**	**86.40**	**76.06**	**85.14**
#09	rotation = 270°	91.49	84.32	91.04	84.73	73.51	83.33
(e) Number of PMMs in Each PMM Block Across Decoder Stages							
#10	{1,1,1,1,1}	91.42	84.19	90.96	86.01	75.46	84.71
#11	{2,2,2,2,1}	91.62	84.54	91.18	86.12	75.63	84.86
#12	{2,3,2,1,1}	**91.74**	**84.75**	**91.31**	**86.40**	**76.06**	**85.14**
#13	{3,3,3,1,1}	91.45	84.25	91.00	86.20	75.74	84.96
(f) Pyramid Branch							
#14	w/o multi-scale local context branch	91.49	84.31	91.04	86.17	75.70	84.95

model complexity and performance. The contribution of the multiscale local context branch in the PMM decoder was further examined by removing this pyramid structure and retaining only the main directional decoding stream (row #14 of Table VI). Without this component, the F1/IoU/Kappa metrics decreased by 0.25%/0.44%/0.27% on the LEVIR-CD and 0.23%/0.36%/0.19% on GZ-CD. These results indicate that the pyramid branch provides essential complementary multiscale cues, enabling the decoder to aggregate fine-grained local patterns and global contextual semantics more effectively. Its absence weakens spatial coherence in boundary regions, confirming that hierarchical feature fusion across scales is vital for precise change delineation.

5) Analysis of Image Size and Performance: To further analyze how LGMM-Net scales with increasing image size and scene coverage, we conduct a systematic scalability study on LEVIR-CD using four input resolutions, 128×128, 256×256, 512×512, and the original unpatched 1024×1024. Fig. 13 summarizes both accuracy and efficiency metrics, including F1, inference latency (Infer Time), throughput (FPS), GPU memory usage (VRAM), computational cost (FLOPs), and the number of parameters (Params). Since the network architecture is unchanged across resolutions, the parameter count remains constant 15.18 M. In contrast, computation and memory scale rapidly with input size: FLOPs increase from 7.72 to 30.88, 123.52, and 494.09 G, while VRAM rises from 758 to 904, 2410, and 8158 MB. These increases translate into substantially higher latency and lower throughput, with inference time growing from 111.7 to 160.0, 377.9, and 1312.7 ms, and FPS dropping from 8.95 to 6.25, 2.65, and 0.76, respectively.

In terms of accuracy, the F1 improves from 90.99 at 128×128 to 91.74 at 256×256, and peaks at 91.89 at 512×512. The gain is mainly attributed to richer spatial context and more complete object structures, which help the model capture long-range dependencies while retaining fine-grained details. However, increasing the resolution to 1024×1024 provides only a marginal benefit and slightly degrades performance to 91.85. We attribute this to the introduction of large amounts of irrelevant background and stronger intrascene variability, which can dilute change cues and make optimization and inference less focused, while dramatically increasing computational cost and latency.

Overall, although 512×512 achieves the highest F1, 256×256 offers the best tradeoff between accuracy and efficiency under our current experimental setup, providing competitive F1 with substantially lower FLOPs, memory footprint, and latency. We therefore adopt 256×256 as the default resolution for the following experiments, which is more suitable for large-area or near–real–time applications.

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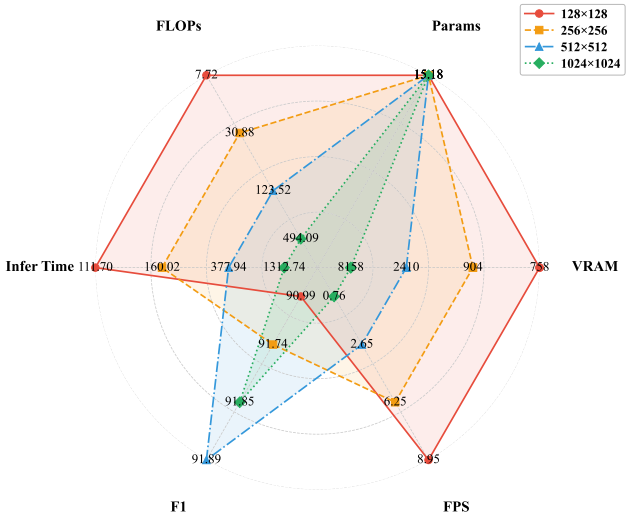


Fig. 13. Accuracy and computational cost under different input resolutions. Infer time denotes inference latency (ms), FPS denotes throughput (frames per second), VRAM denotes GPU memory usage during inference (MB), Params denotes the number of parameters (M), and FLOPs denotes computational complexity (G). F1 is the F1 on the LEVIR-CD dataset. All results are obtained using a single model trained at 256×256, and evaluated at 128×128, 256×256, 512×512, and 1024×1024.

Summary of Key Findings: The ablation studies validate the effectiveness of our core design choices as follows.

1) *DLGPE:* Enables parallel modeling of local details and global semantics within a unified encoder through hierarchical feature extraction, providing complementary enhancement that improves feature discriminability.
2) *Soft Channel Splitting:* Adaptively balances local and global information flowthrough dynamic feature allocation, outperforming hard splitting by allowing flexible capacity distribution between parallel branches.
3) *PMM:* Integrates direction-aware long-range dependency modeling with multiscale local feature extraction in a unified decoder, enabling explicit change boundary reconstruction through parallel Mamba and pyramid pathways.
4) *MGD-SS2D:* Leverages directional masks to guide efficient four-directional scanning, enhancing anisotropic object representation and boundary preservation while reducing computational overhead compared to multidirectional alternatives.

F. Heatmap Analysis

To provide an intuitive understanding of how LGMM-Net operates, we visualize multistage features across the encoder–decoder pipeline, as shown in Fig. 14. In the encoder, feature resolutions are progressively reduced from 64×64 to 32×32, 16×16, and 8×8. The decoder then upsamples features step by step, restoring them to 256×256. For consistency, all heatmaps are resized to 256×256 for visualization.

At the shallow stages (64×64 and 32×32), activations are mainly concentrated along local edges and texture regions. This indicates that the model effectively captures fine-grained structural information, but its perception of global change patterns remains limited. As the resolution decreases to 16×16 and 8×8, the deeper features exhibit stronger spatial consistency and global semantics. This enables the model to distinguish large-scale variations between bi-temporal images, although boundary localization becomes less precise.

In the decoder, low-resolution global semantics are gradually fused with high-resolution details. At 64×64, 128×128, and finally 256×256, activations increasingly align with the boundaries of buildings and man-made structures, closely matching the true change regions. This hierarchical feature reconstruction demonstrates that LGMM-Net not only preserves global contextual understanding but also restores spatial details, enabling precise localization of change regions.

Overall, these visualizations clearly illustrate the layer-wise feature evolution of LGMM-Net. The results are consistent with the qualitative change maps reported earlier, further validating the effectiveness and robustness of the proposed method.

V. Limitations and Future Work

As reported in Table I, our method demonstrates robust performance on conventional benchmarks, achieving an F1 of 91.74% and IoU of 84.75% on LEVIR-CD, 97.71% and 95.51% on CDD, and 86.40% and 76.06% on GZ-CD. In contrast, a noticeable performance gap is observed on the UAV-CD+ (see Table II), where the F1 and IoU scores decrease to 70.10% and 53.96%, respectively.

UAV-CD+ is characterized by ultrahigh-resolution imagery containing extremely fine-grained changes, many of which are isolated and lack distinct structural boundaries. We attribute this performance drop to the specific inductive biases of our core modules, DLGPE and PMM, which prioritize structural and global semantic consistency. Specifically, DLGPE is designed to capture geometric structures for feature alignment, but extremely fine-grained changes often appear as isolated pixels lacking clear geometric patterns, making them difficult to encode effectively. Furthermore, PMM focuses on direction-aware long-range dependency modeling to suppress pseudo-changes, such as seasonal differences. While this mechanism ensures the semantic integrity of objects, it may inadvertently filter out nonstructural, pixel-level variations that resemble noise.

As discussed in Section IV-D, LGMM-Net shows advantages in terms of parameter count and computational cost compared with several competing methods, yet its end-to-end inference throughput still leaves clear room for improvement. Under identical testing settings (256×256 patches, a single NVIDIA A40 GPU, and the same evaluation conditions), our method achieves an inference time of 160.02 ms/image and a throughput of 6.25 images/second. For reference, ELGC-Net (our baseline) achieves 54.35 ms/image and 18.40 images/second, whereas ChangeMamba-B achieves 91.31 ms/image and 10.73 images/second (see Table III). These results indicate that practical runtime is not determined solely by FLOPs, but is also influenced by operator implementations, memory-access overhead, and sequential dependencies. In particular, PMM relies on direction-aware

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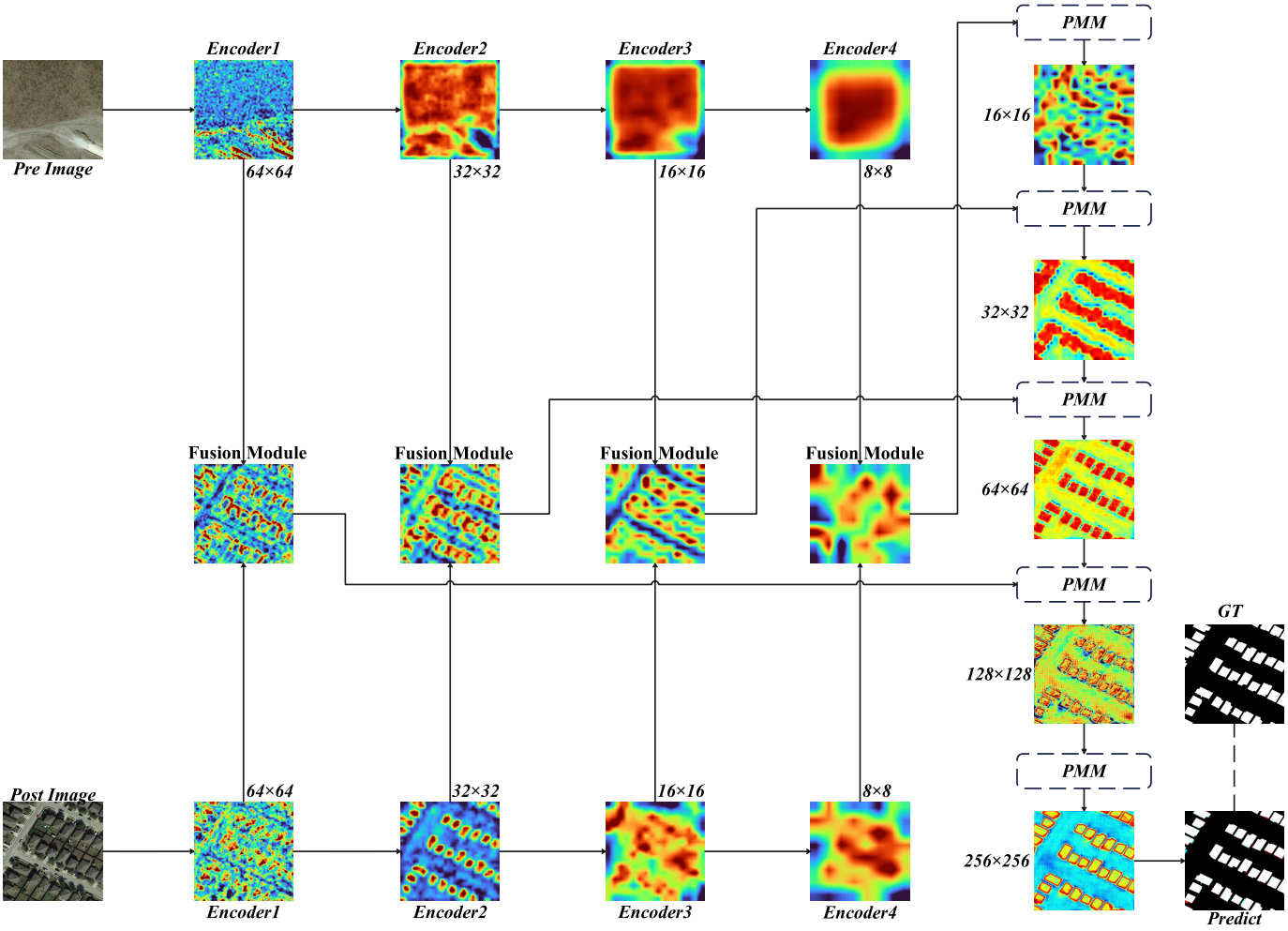


Fig. 14. Heatmap visualizations of multistage features in LGMM-Net from LEVIR-CD dataset. Early layers focus on local textures, while deeper layers encode global context. The decoder progressively restores fine boundaries, highlighting the network's ability to localize true changes. Color map: red/yellow indicates strong activation or high change likelihood, green indicates moderate response, and blue indicates low response/background.

state-space scanning and mask-guided feature routing. While this design strengthens global modeling and robustness, it may introduce sequential computation and additional memory traffic, thereby limiting practical throughput. To improve deployability, we plan to enhance efficiency from three perspectives:

1) *Lightweight Variants:* We will explore lighter PMM designs, e.g., reducing decoder depth, sharing the MGD-SS2D scan parameters (including masked directional kernels and state-space projections) across decoder stages, limiting the number of scan directions, and replacing some high-resolution PMM blocks with depth-wise separable convolutions to alleviate sequential computation and memory pressure.
2) *Structured Pruning and Distillation:* We will investigate structured pruning at the channel and block levels, and combine it with knowledge distillation to train smaller student models, improving inference speed while largely preserving accuracy.
3) *Engineering Acceleration:* We will adopt kernel fusion for scan-related operators, mixed precision or INT8 quantization, and compiler- and runtime-level graph optimizations and operator acceleration, so that theoretical efficiency can translate more effectively into practical throughput gains.

Meanwhile, we will further advance this line of work in two directions. First, we will explore frequency-aware representation learning to disentangle high-frequency change signals from background noise in the spectral domain, which may reduce noise interference in fine-grained CD. We will also leverage multimodal vision–language models (VLMs) to introduce text-guided semantic priors, providing higher level contextual constraints for localizing ambiguous fine-grained changes. Second, we will improve deployability by extending LGMM-Net to cross-sensor and semi-supervised settings to better handle domain shifts and limited annotations. We will further investigate SNN-inspired mechanisms for higher efficiency and robust real-time CD under resource constraints, including on-device unsupervised adaptation [44], spatiotemporal pruning for ultralow-latency training [45], and self-supervised high-order information bottlenecks for robust event-based modeling [46].

VI. Conclusion

In this article, we proposed LGMM-Net, a novel hybrid framework for RS CD that introduces coordinated innovations

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in both the encoder and decoder. First, we designed a soft-split strategy that assigns features to parallel branches-multiconv, fw_attn, and sw_attn. This design enables simultaneous modeling of local details and global semantics within the same layer, thereby enhancing the representation of complex geospatial scenes. Second, we design the PMM as a direction-aware multiscale decoder block. It integrates the MGD-SS2D, which employs directional convolutions and feature rotations for efficient four-directional scanning, with multibranch dilated convolutions that aggregate pyramid features to capture multiscale local context and preserve fine boundaries. Through this design, PMM effectively suppresses pseudo-changes, restores anisotropic objects, and enhances boundary completeness, while maintaining computational efficiency. By coupling adaptive enhancement in the encoder with cross-scale, direction-aware decoding, LGMM-Net maintains reliability under illumination/seasonal/sensor variability, effectively reducing pseudo-changes without harming recall. Extensive experiments on four public datasets LEVIR-CD, CDD, GZ-CD, and UAV-CD+ demonstrate that LGMM-Net achieves SOTA performance in quantitative evaluation, qualitative visualization, and ablation analysis. The results show that our method not only improves detection accuracy but also provides clear advantages in suppressing false alarms and reducing missed-detection regions. Furthermore, heatmap analyses qualitatively confirm the effectiveness of LGMM-Net in multiscale feature fusion and precise identification of change regions.

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